Comparison Between RS67 Vs RS57-70

Chatura Kuruppu

April 22, 2026

Input Files Used

```
/seaquest/users/harshaka/e906_project/e906-root-ana/work_gpvm/scripts/results_runFinalTree/merged_results_e906_mixing/rs57/merged_RS57_LH2_1
/seaquest/users/harshaka/e906_project/e906-root-ana/work_gpvm/scripts/results_runFinalTree/merged_results_e906_mixing/rs57/merged_RS57_LD2_3
/seaquest/users/harshaka/e906_project/e906-root-ana/work_gpvm/scripts/results_runFinalTree/merged_results_e906_mixing/rs57/merged_RS57_Empty
/seaquest/users/harshaka/e906_project/e906-root-ana/work_gpvm/scripts/results_runFinalTree/merged_results_e906_mixing/rs59/merged_RS59_LH2_1
/seaquest/users/harshaka/e906_project/e906-root-ana/work_gpvm/scripts/results_runFinalTree/merged_results_e906_mixing/rs59/merged_RS59_LD2_3
/seaquest/users/harshaka/e906_project/e906-root-ana/work_gpvm/scripts/results_runFinalTree/merged_results_e906_mixing/rs59/merged_RS59_Empty
/seaquest/users/harshaka/e906_project/e906-root-ana/work_gpvm/scripts/results_runFinalTree/merged_results_e906_mixing/rs62/merged_RS62_LH2_1
/seaquest/users/harshaka/e906_project/e906-root-ana/work_gpvm/scripts/results_runFinalTree/merged_results_e906_mixing/rs62/merged_RS62_LD2_3
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/seaquest/users/harshaka/e906_project/e906-root-ana/work_gpvm/scripts/results_runFinalTree/merged_results_e906_mixing/rs70/merged_RS70_Empty
```

Remark: Currently we don't process RS5a, RS5b files.

Analysis Constants (config.py)

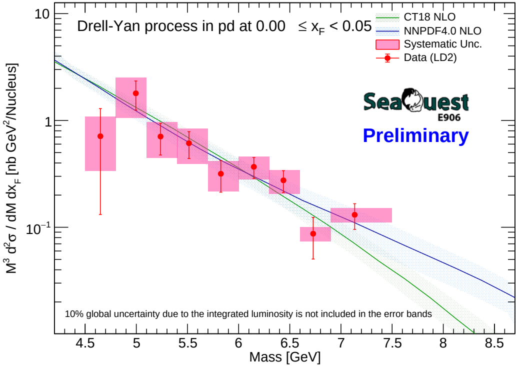
```
INPUT_NPZ_FILE = "interpolation_data_d1.npz"

PROTONS_ON_TARGET_LH2 = 2.763911e+17
PROTONS_ON_TARGET_LD2 = 1.319723e+17
PROTONS_ON_TARGET_FLASK = 5.590528e+16

LH2_TARGET_DENSITY_MOL_CM2 = 3.5966
LD2_TARGET_DENSITY_MOL_CM2 = 8.0431

LH2_TARGET_LENGTH_CM = 50.8
LD2_TARGET_LENGTH_CM = 50.8

AVOGADRO_CONSTANT = 6.022e23
XF_BIN_WIDTH = 0.05
# ... (Derived Normalizations Continue as configured in source...)
```

Event Selection Used

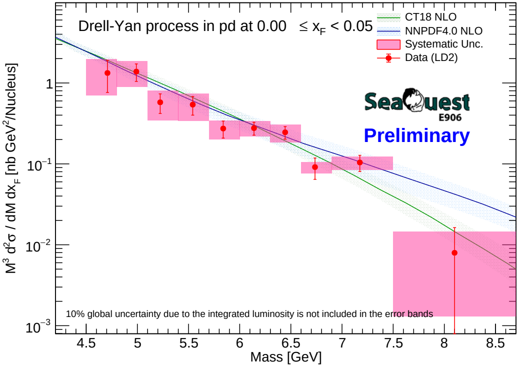
```
def get_e906_chuck_cuts_string(cut_val=4.2):
    bo = "(runID >= 11000 ? 1.6 : 0.4)"

    dimuon_cut = (
        f"(abs(dx) < 0.25) && (abs(dy - {bo}) < 0.22) && "
        f"(dz < -5.) && (dz > -280.) && (abs(dpx) < 1.8) && (abs(dpy) < 2.0) && "
        f"(dpx * dpx + dpy * dpy < 5.) && (dpz < 116.) && (dpz > 38.) && "
        f"(mass > {cut_val}) && (mass < 8.8) && "
        f"(dx * dx + (dy - {bo}) * (dy - {bo}) < 0.06) && "
        f"(xF < 0.95) && (xF > -0.1) && (xT > 0.05) && (xT <= 0.58) && "
        f"(abs(costh) < 0.5) && (abs(trackSeparation) < 270.) && "
        f"(chisq_dimuon < 18)"
    )
    # ... track cuts continue
```

CrossSection_LD2_xF_0.00_0.05_Centroid_with_logo.pdf

RS67

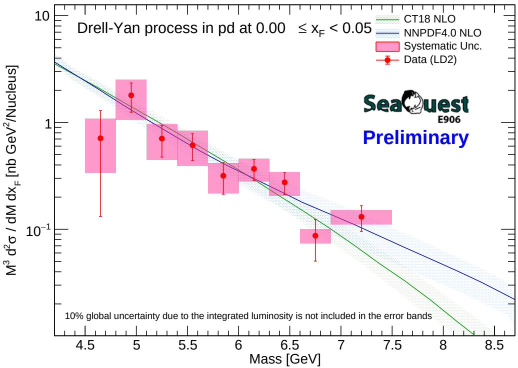


RS57-70

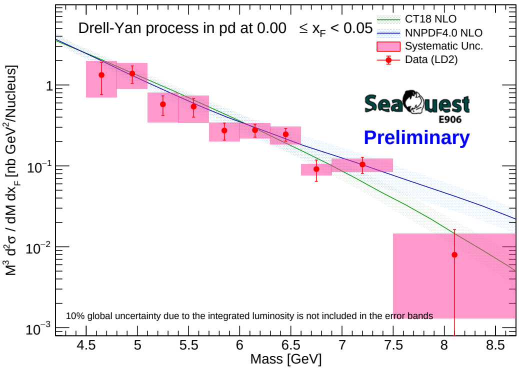


CrossSection_LD2_xF_0.00_0.05_GeoCenter_with_logo.pdf

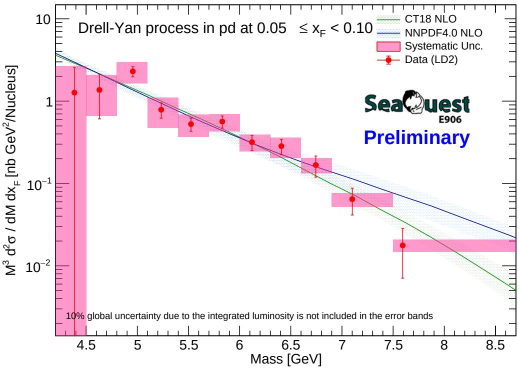
RS67



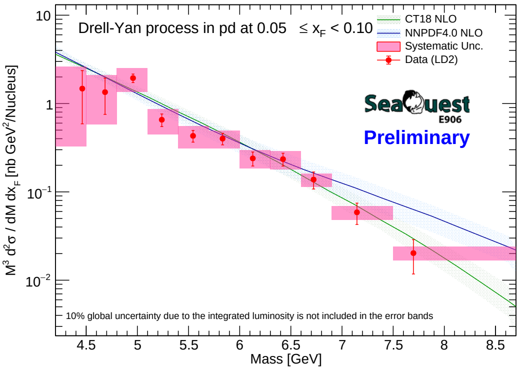
RS57-70



RS67


Drell-Yan process in pd at 0.05 $\leq x_F < 0.10$
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\, d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]
10
1
10^{-1}
10^{-2}
4.5
5
5.5
6
6.5
7
7.5
8
8.5

RS57-70


Drell-Yan process in pd at 0.05 $\leq x_F < 0.10$
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest
E906
Preliminary
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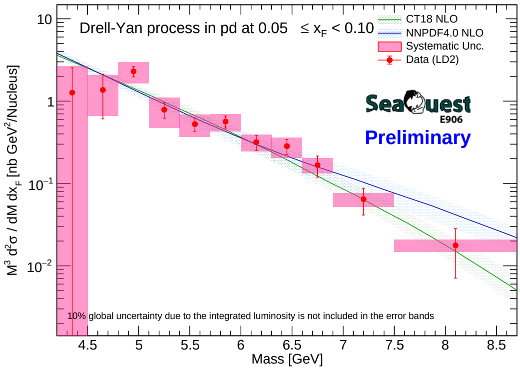
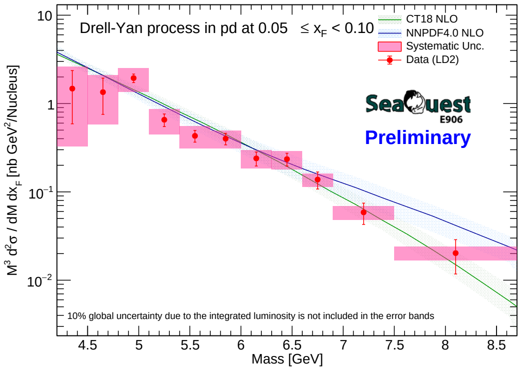
Mass [GeV]
10
1
10^{-1}
10^{-2}
4.5
5
5.5
6
6.5
7
7.5
8
8.5

CrossSection_LD2_xF_0.05_0.10_GeoCenter_with_logo.pdf

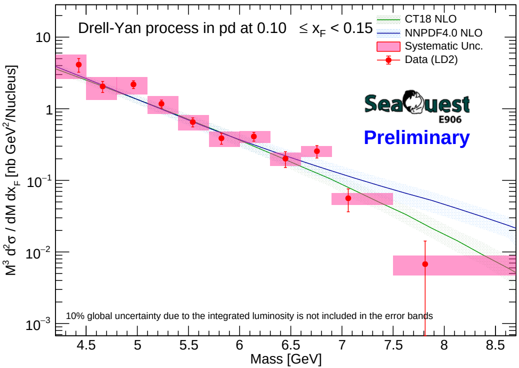
RS67



RS57-70

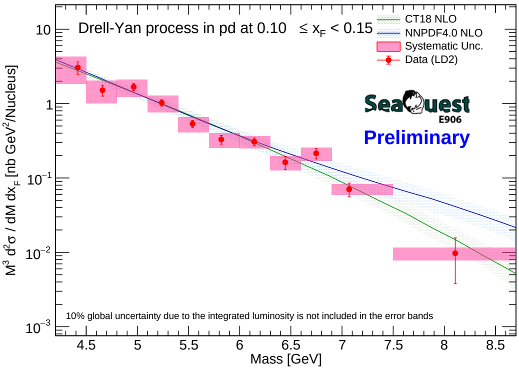


RS67



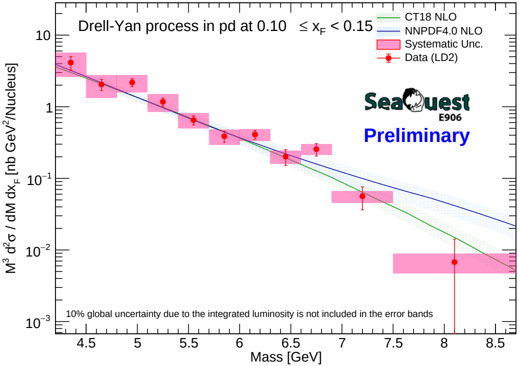
Drell-Yan process in pd at 0.10 ≤ x_F < 0.15
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70

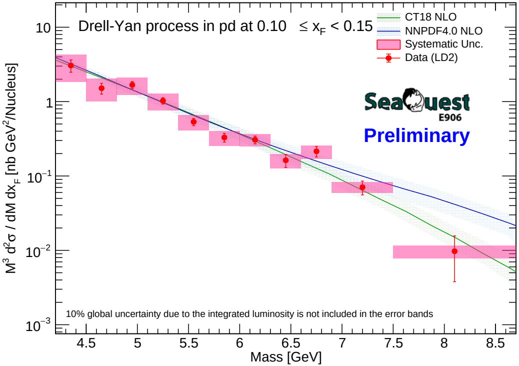


Drell-Yan process in pd at 0.10 ≤ x_F < 0.15
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS67

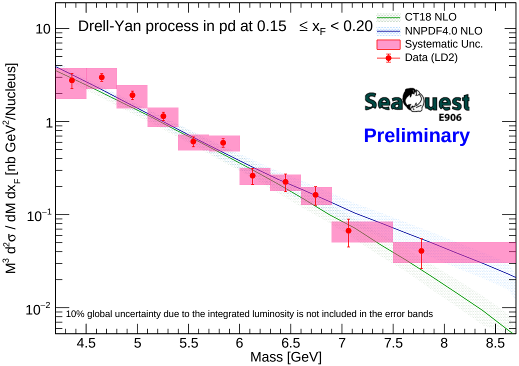

Drell-Yan process in pd at 0.10 ≤ x_F < 0.15
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70

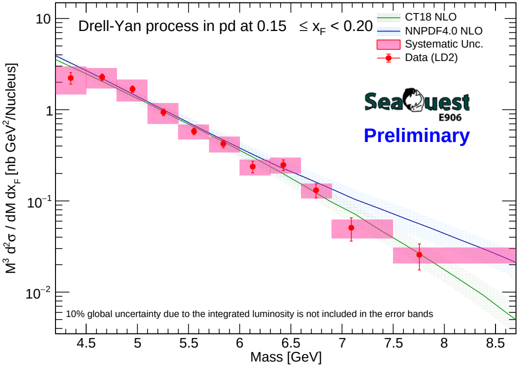

Drell-Yan process in pd at 0.10 ≤ x_F < 0.15
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

CrossSection_LD2_xF_0.15_0.20_Centroid_with_logo.pdf

RS67

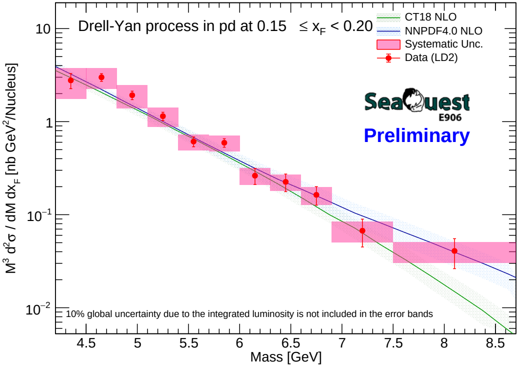


RS57-70



CrossSection_LD2_xF_0.15_0.20_GeoCenter_with_logo.pdf

RS67



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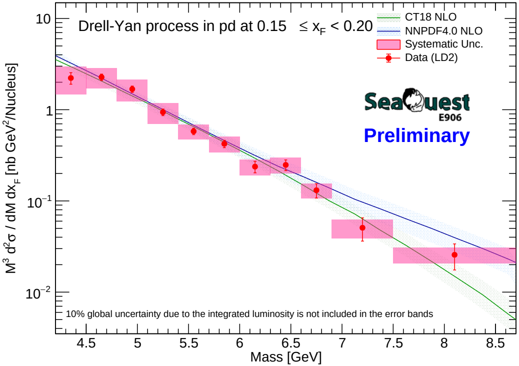
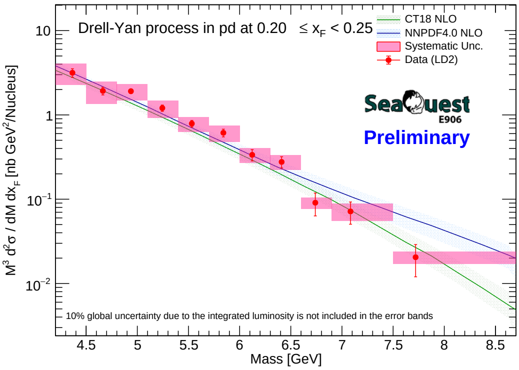
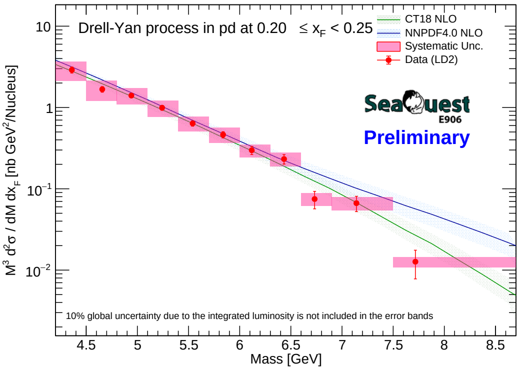


RS67

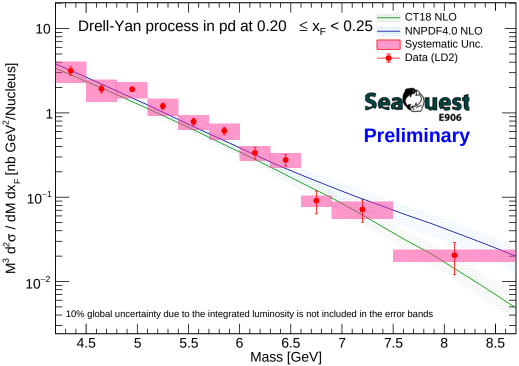

Drell-Yan process in pd at 0.20 ≤ x_F < 0.25
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\, d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70

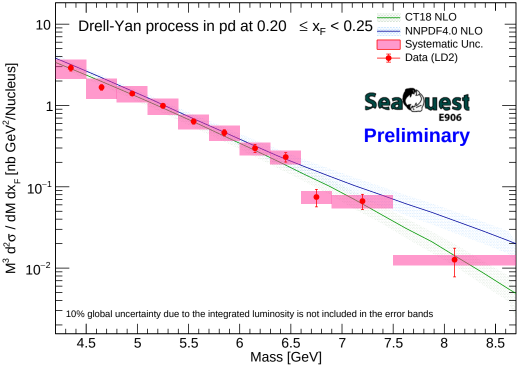

Drell-Yan process in pd at 0.20 ≤ x_F < 0.25
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\, d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

CrossSection_LD2_xF_0.20_0.25_GeoCenter_with_logo.pdf

RS67

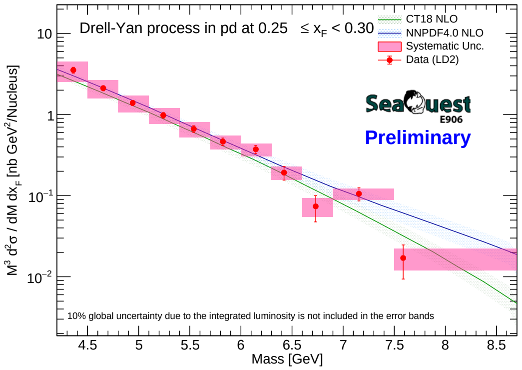


RS57-70

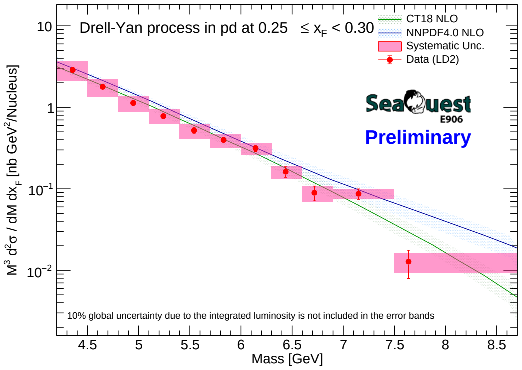


CrossSection_LD2_xF_0.25_0.30_Centroid_with_logo.pdf

RS67

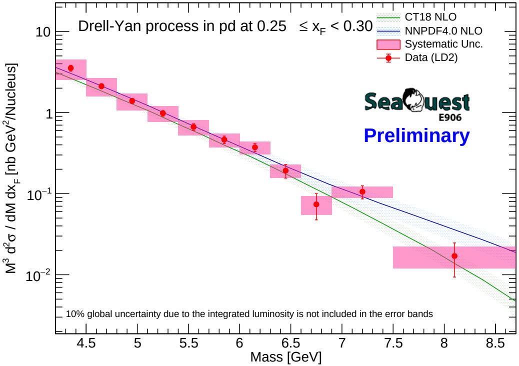


RS57-70



CrossSection_LD2_xF_0.25_0.30_GeoCenter_with_logo.pdf

RS67



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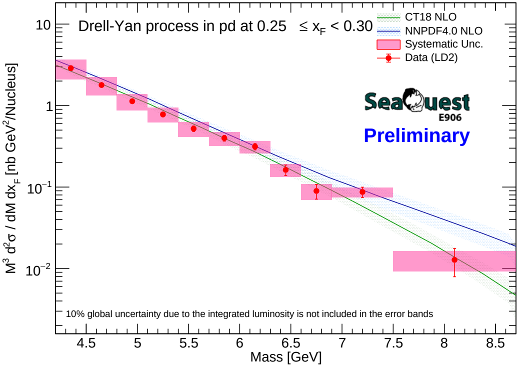
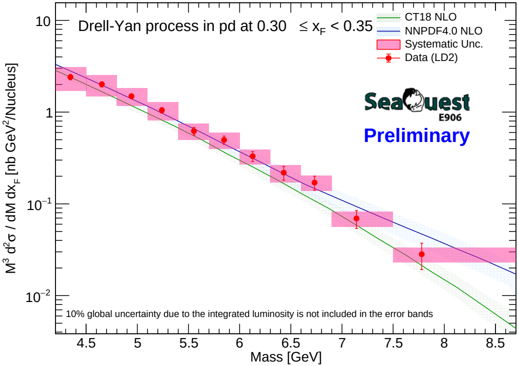
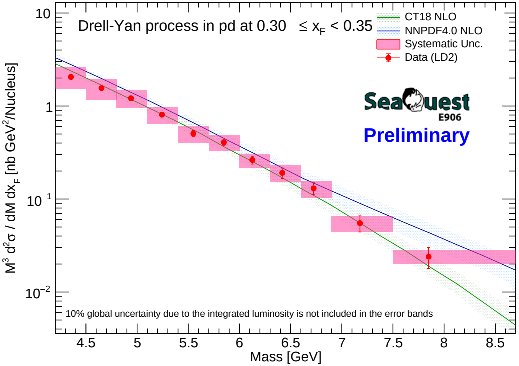


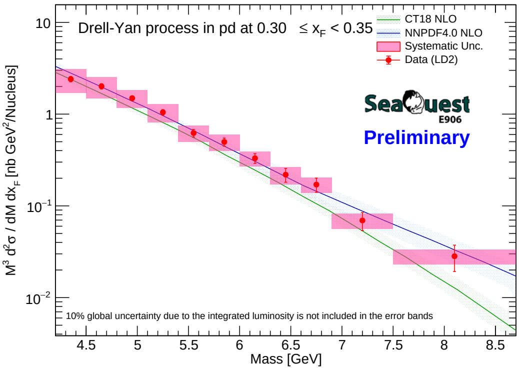
RS67


Drell-Yan process in pd at 0.30 ≤ x_F < 0.35
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]
10
1
10^{-1}
10^{-2}
4.5
5
5.5
6
6.5
7
7.5
8
8.5

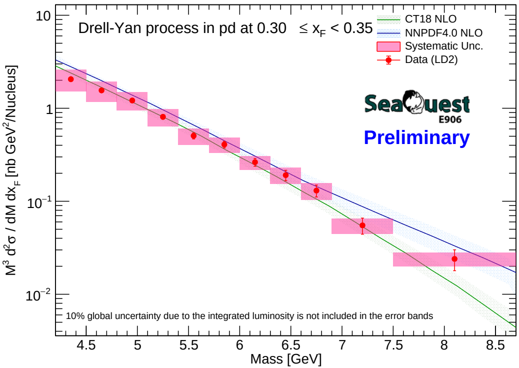
RS57-70


Drell-Yan process in pd at 0.30 ≤ x_F < 0.35
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]
10
1
10^{-1}
10^{-2}
4.5
5
5.5
6
6.5
7
7.5
8
8.5

RS67

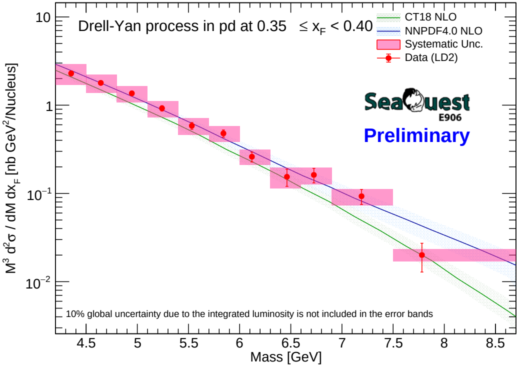

Drell-Yan process in pd at 0.30 ≤ x_F < 0.35
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70


Drell-Yan process in pd at 0.30 ≤ x_F < 0.35
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

CrossSection_LD2_xF_0.35_0.40_Centroid_with_logo.pdf

RS67



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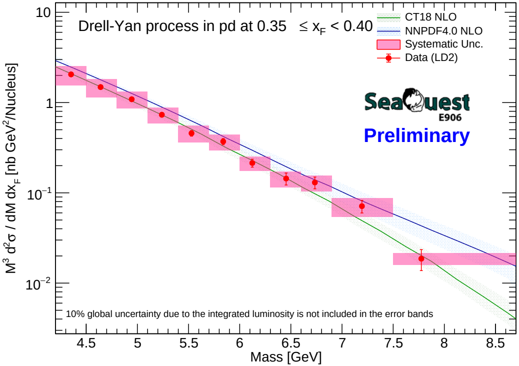
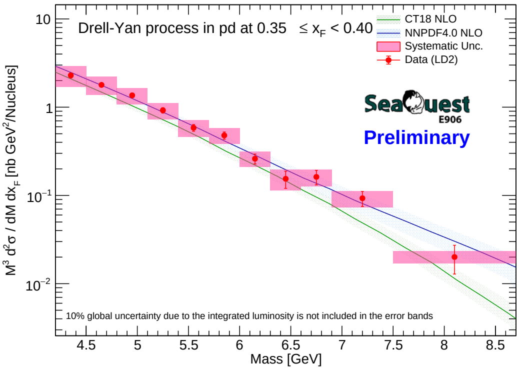
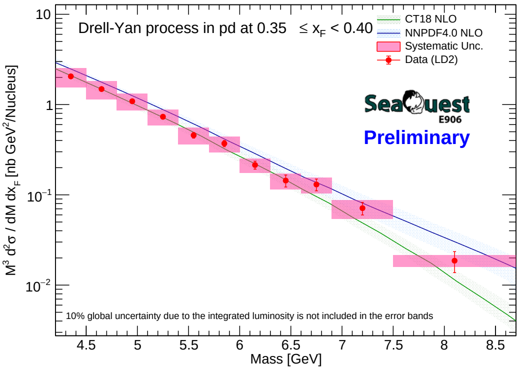


RS67

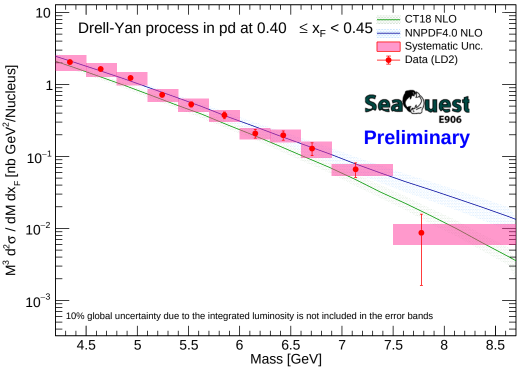

Drell-Yan process in pd at 0.35 $\leq x_F < 0.40$
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\, d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70

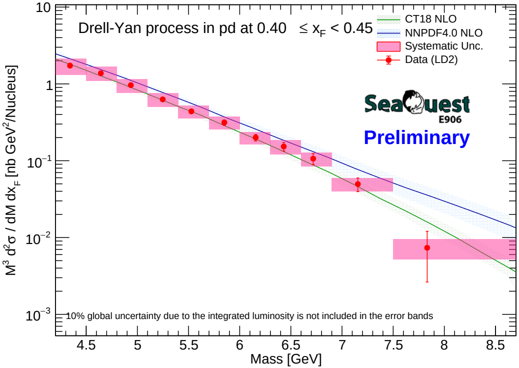

Drell-Yan process in pd at 0.35 $\leq x_F < 0.40$
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\, d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

CrossSection_LD2_xF_0.40_0.45_Centroid_with_logo.pdf

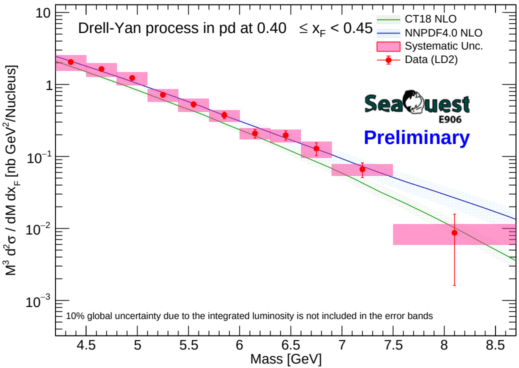
RS67



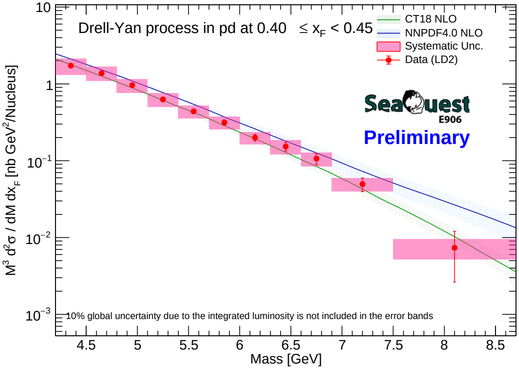
RS57-70



RS67

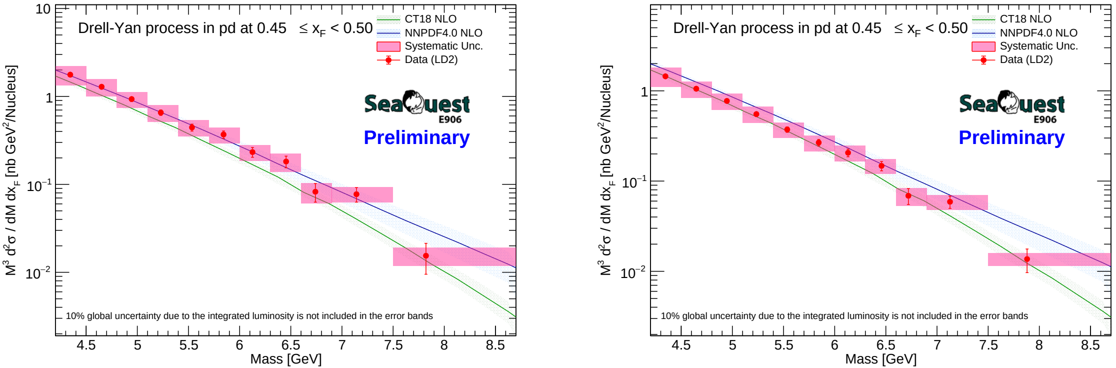

Drell-Yan process in pd at 0.40 ≤ x_F < 0.45
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70

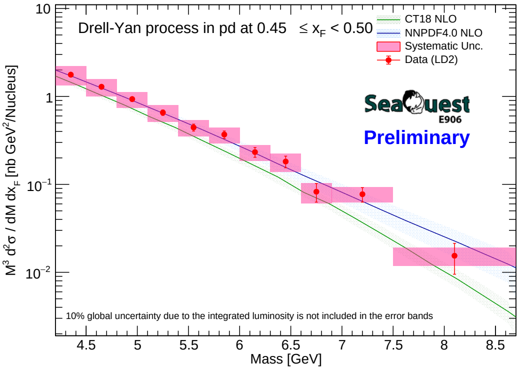

Drell-Yan process in pd at 0.40 ≤ x_F < 0.45
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

CrossSection_LD2_xF_0.45_0.50_Centroid_with_logo.pdf

RS67 **RS57-70**

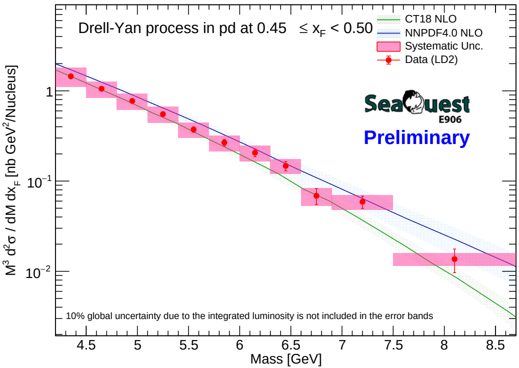


RS67



Drell-Yan process in pd at 0.45 ≤ x_F < 0.50
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

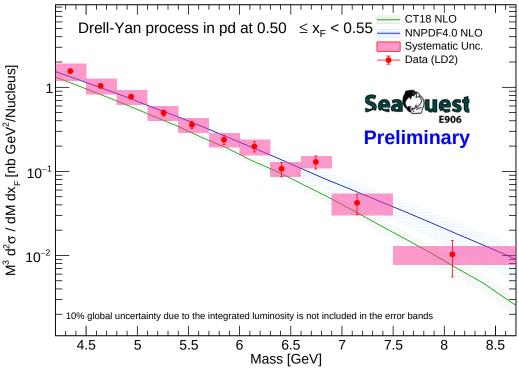
RS57-70



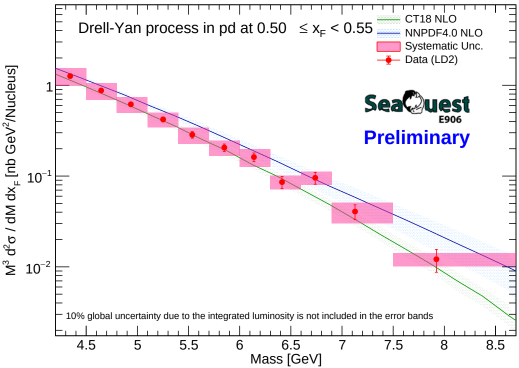
Drell-Yan process in pd at 0.45 ≤ x_F < 0.50
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

CrossSection_LD2_xF_0.50_0.55_Centroid_with_logo.pdf

RS67

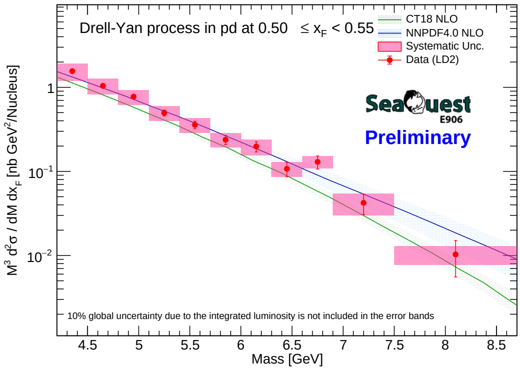


RS57-70

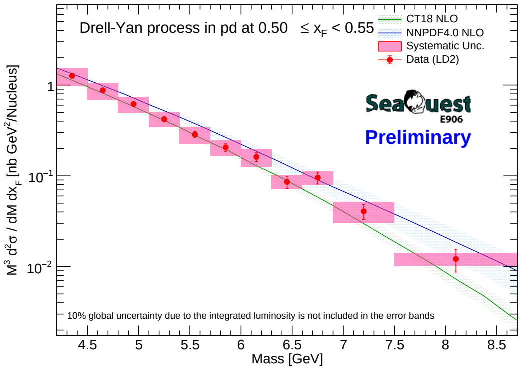


CrossSection_LD2_xF_0.50_0.55_GeoCenter_with_logo.pdf

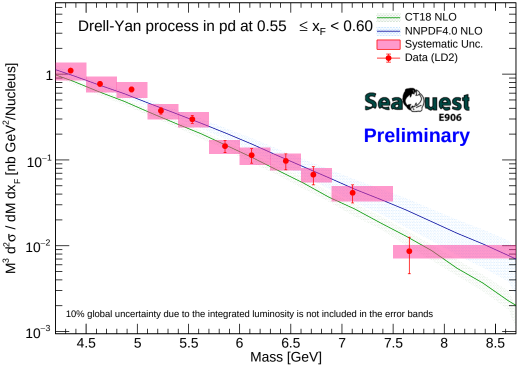
RS67



RS57-70

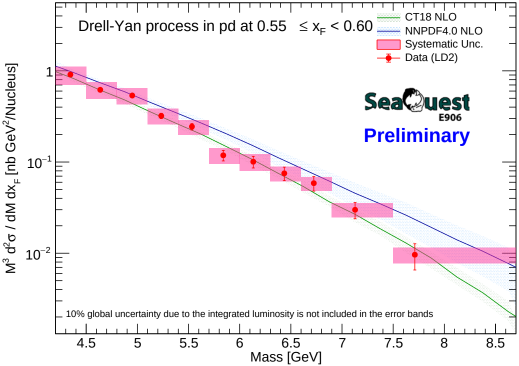


RS67



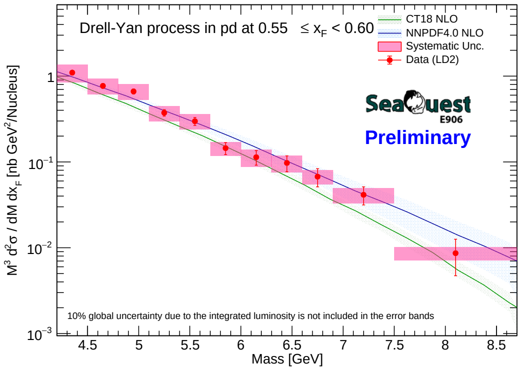
Drell-Yan process in pd at 0.55 ≤ x_F < 0.60
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70

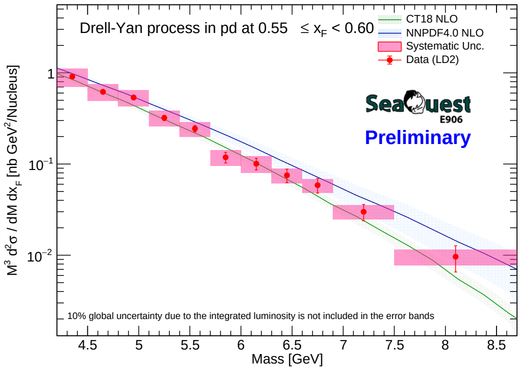


Drell-Yan process in pd at 0.55 ≤ x_F < 0.60
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS67

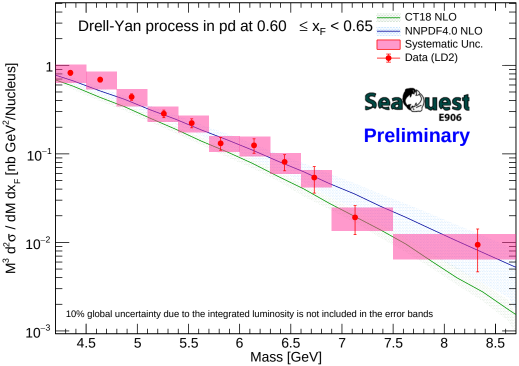

Drell-Yan process in pd at 0.55 ≤ x_F < 0.60
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma\ /\ dM\ dx_F$ [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70

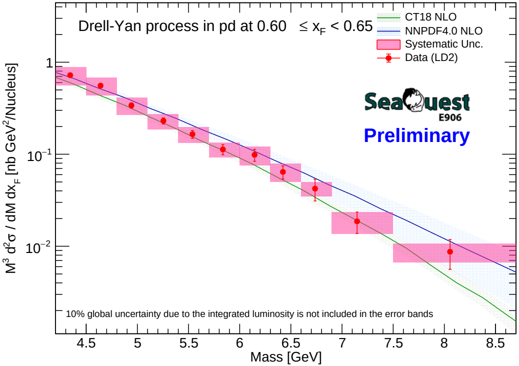

Drell-Yan process in pd at 0.55 ≤ x_F < 0.60
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma\ /\ dM\ dx_F$ [nb GeV^2/Nucleus]
Mass [GeV]

CrossSection_LD2_xF_0.60_0.65_Centroid_with_logo.pdf

RS67

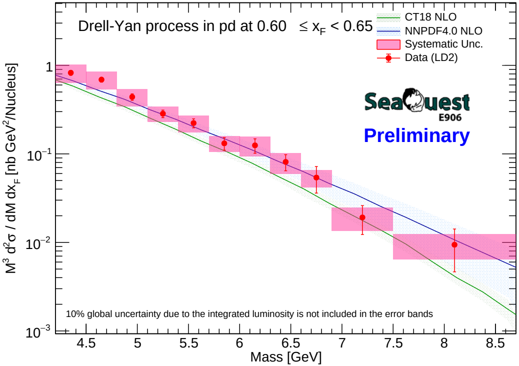


RS57-70

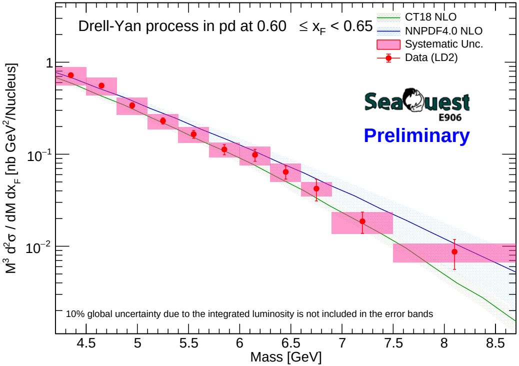


CrossSection_LD2_xF_0.60_0.65_GeoCenter_with_logo.pdf

RS67

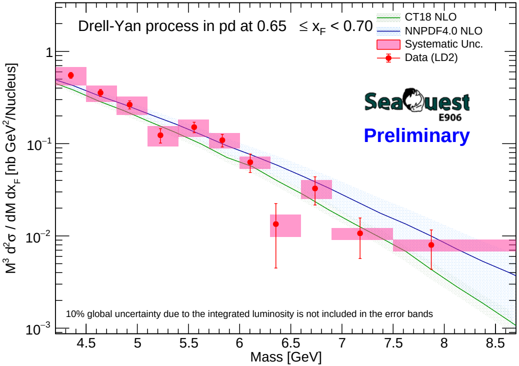


RS57-70

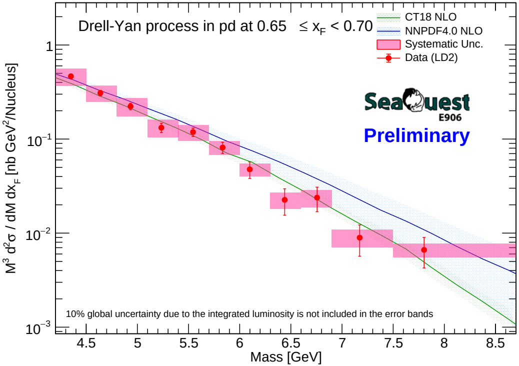


CrossSection_LD2_xF_0.65_0.70_Centroid_with_logo.pdf

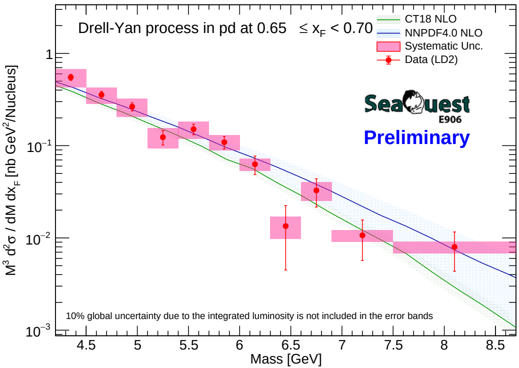
RS67



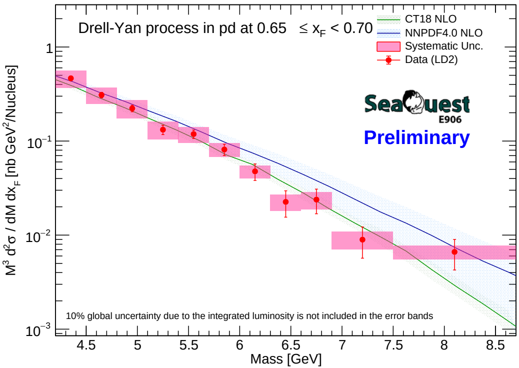
RS57-70



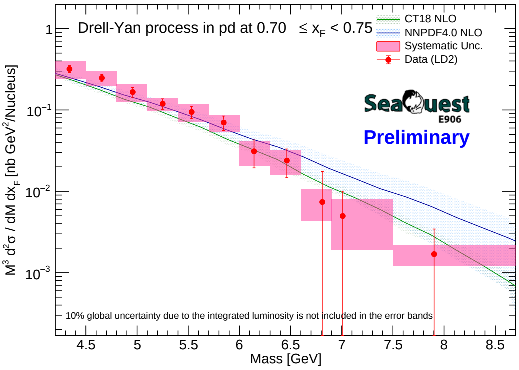
RS67


Drell-Yan process in pd at 0.65 ≤ x_F < 0.70
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]
1
10^{-1}
10^{-2}
10^{-3}
4.5
5
5.5
6
6.5
7
7.5
8
8.5

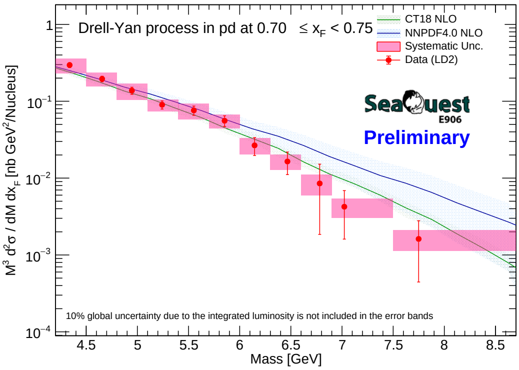
RS57-70


Drell-Yan process in pd at 0.65 ≤ x_F < 0.70
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]
1
10^{-1}
10^{-2}
10^{-3}
4.5
5
5.5
6
6.5
7
7.5
8
8.5

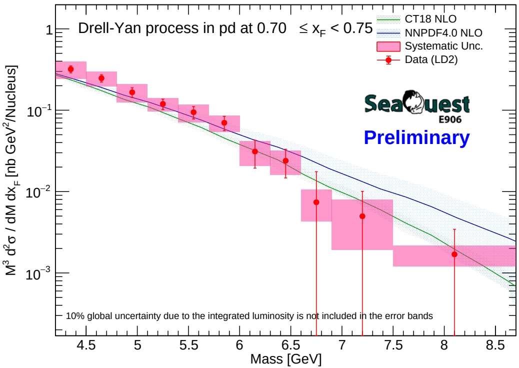
RS67


Drell-Yan process in pd at 0.70 ≤ x_F < 0.75
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
M^3 d^2σ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

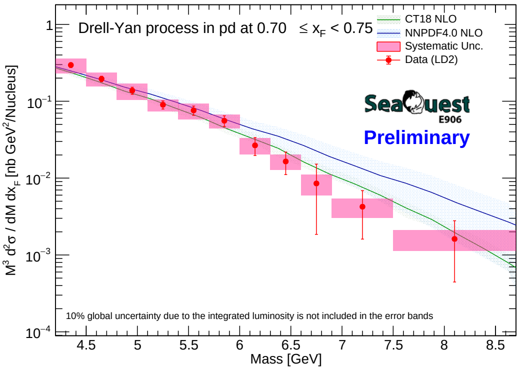
RS57-70


Drell-Yan process in pd at 0.70 ≤ x_F < 0.75
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
M^3 d^2σ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS67

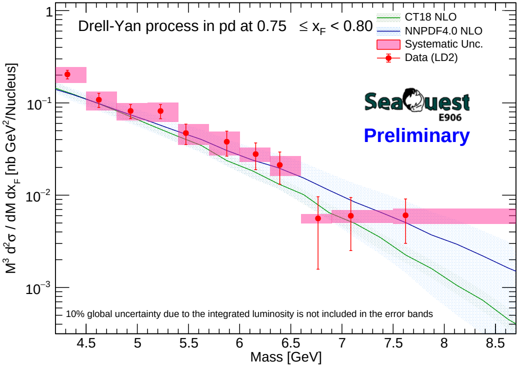

Drell-Yan process in pd at 0.70 ≤ x_F < 0.75
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\, d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70

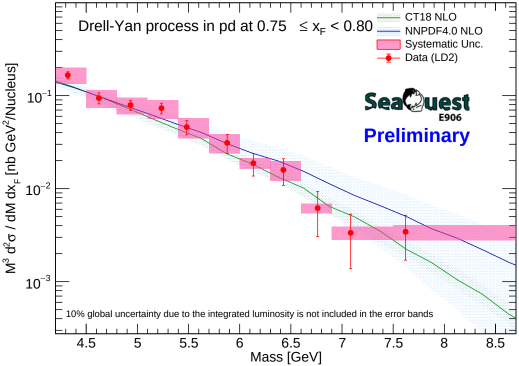

Drell-Yan process in pd at 0.70 ≤ x_F < 0.75
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\, d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

CrossSection_LD2_xF_0.75_0.80_Centroid_with_logo.pdf

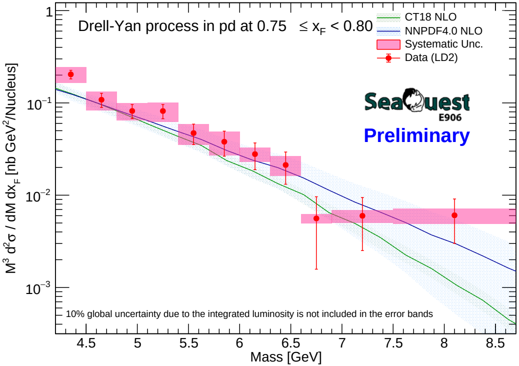
RS67



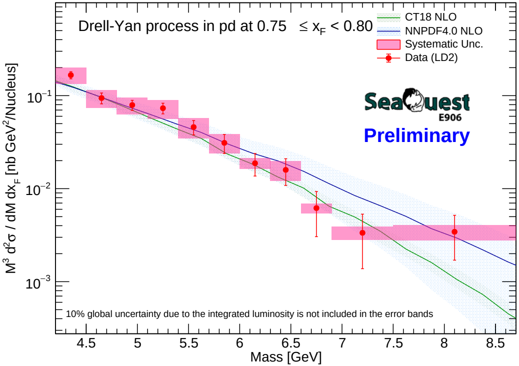
RS57-70



RS67

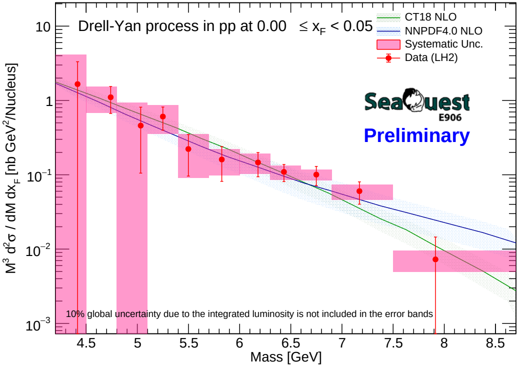

Drell-Yan process in pd at 0.75 ≤ x_F < 0.80
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70

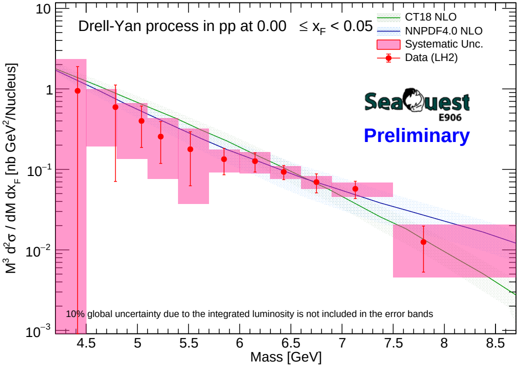

Drell-Yan process in pd at 0.75 ≤ x_F < 0.80
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LD2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

CrossSection_LH2_xF_0.00_0.05_Centroid_with_logo.pdf

RS67



RS57-70



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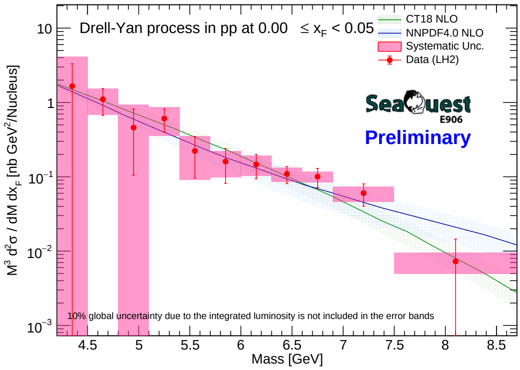
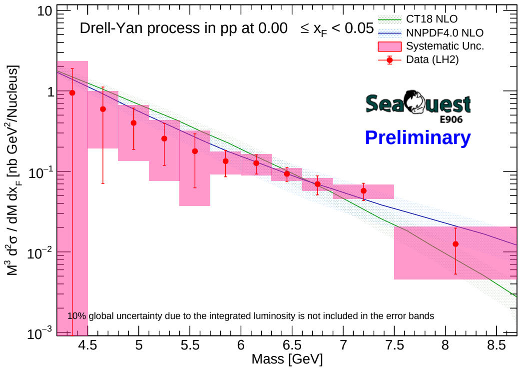
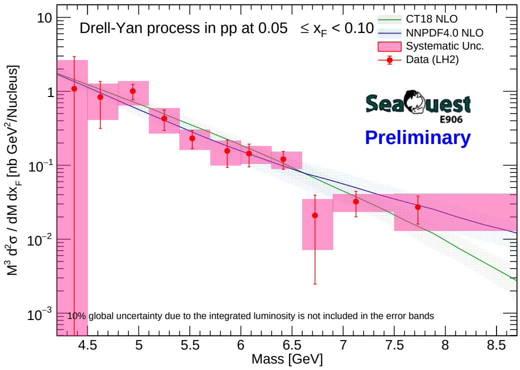

Drell-Yan process in pp at 0.00 ≤ x_F < 0.05
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70

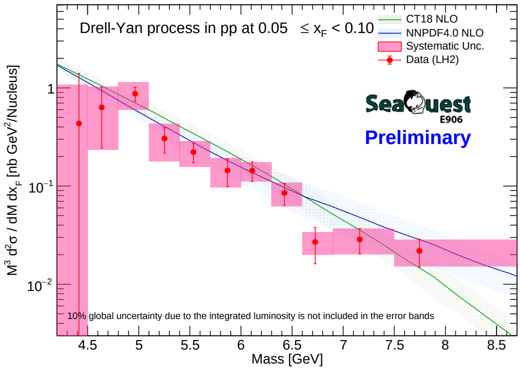

Drell-Yan process in pp at 0.00 ≤ x_F < 0.05
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

CrossSection_LH2_xF_0.05_0.10_Centroid_with_logo.pdf

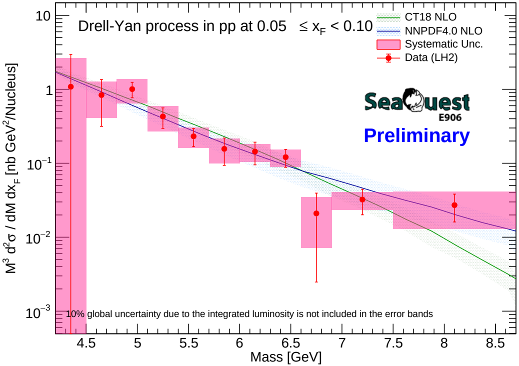
RS67



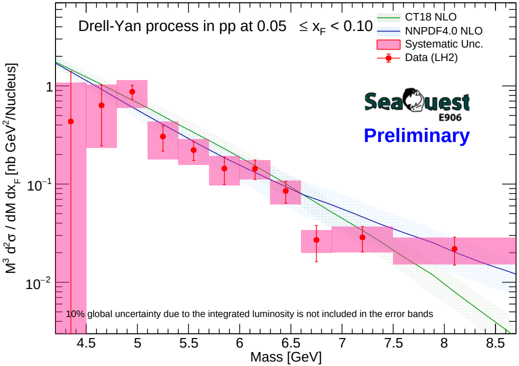
RS57-70



RS67

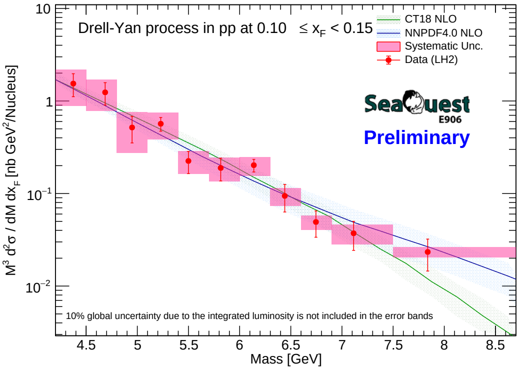

Drell-Yan process in pp at 0.05 ≤ x_F < 0.10
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70

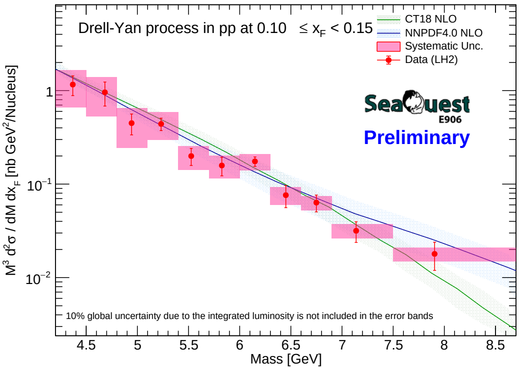

Drell-Yan process in pp at 0.05 ≤ x_F < 0.10
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

CrossSection_LH2_xF_0.10_0.15_Centroid_with_logo.pdf

RS67

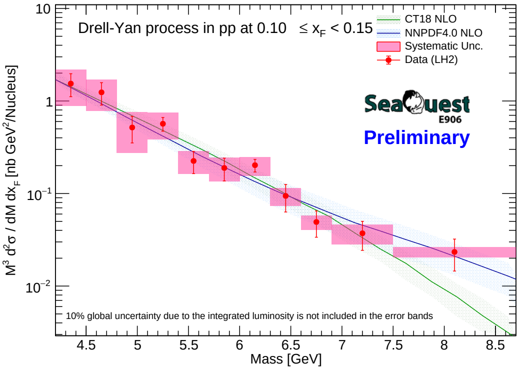


RS57-70

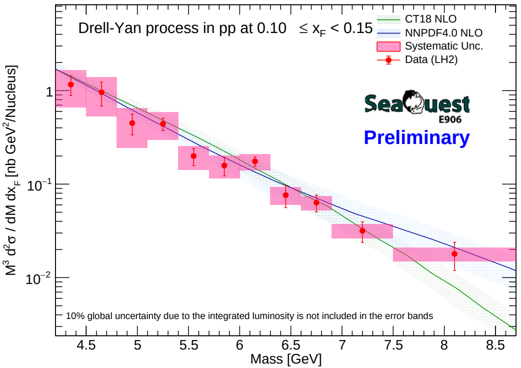


CrossSection_LH2_xF_0.10_0.15_GeoCenter_with_logo.pdf

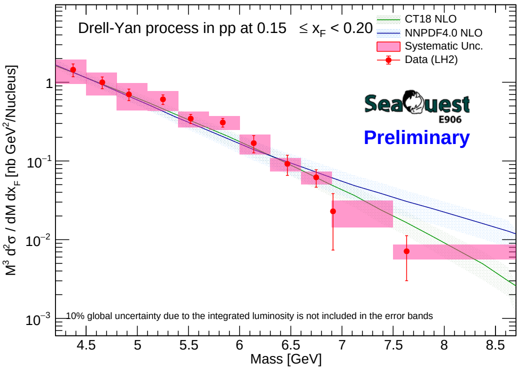
RS67



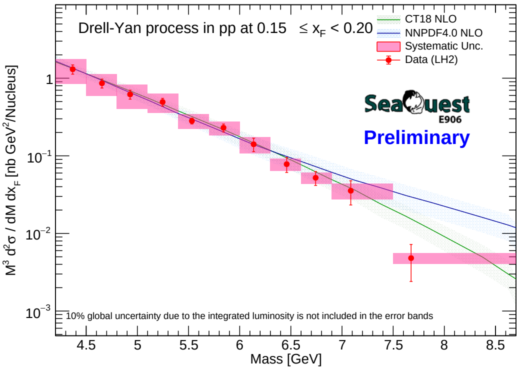
RS57-70



RS67

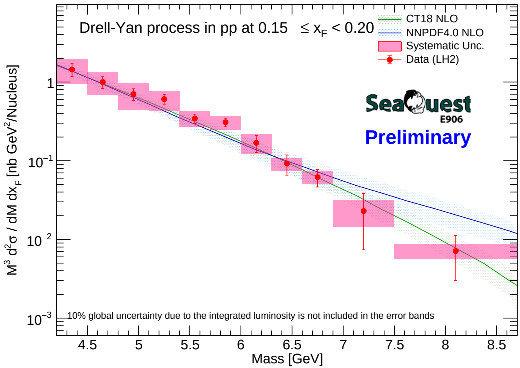

Drell-Yan process in pp at 0.15 ≤ x_F < 0.20
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\, d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70

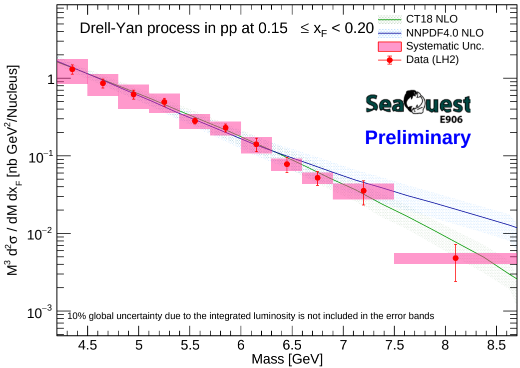

Drell-Yan process in pp at 0.15 ≤ x_F < 0.20
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\, d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

CrossSection_LH2_xF_0.15_0.20_GeoCenter_with_logo.pdf

RS67

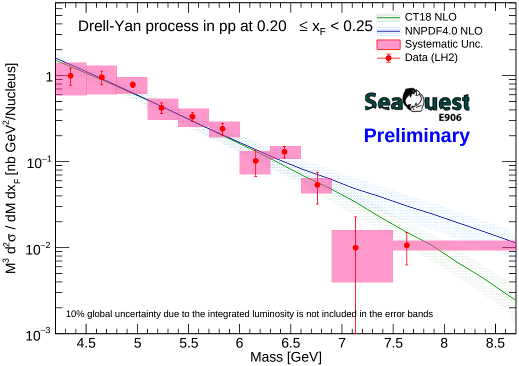


RS57-70

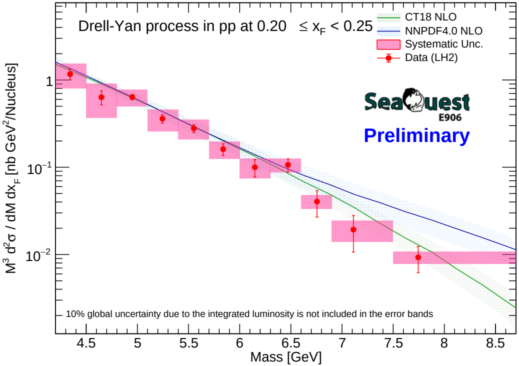


CrossSection_LH2_xF_0.20_0.25_Centroid_with_logo.pdf

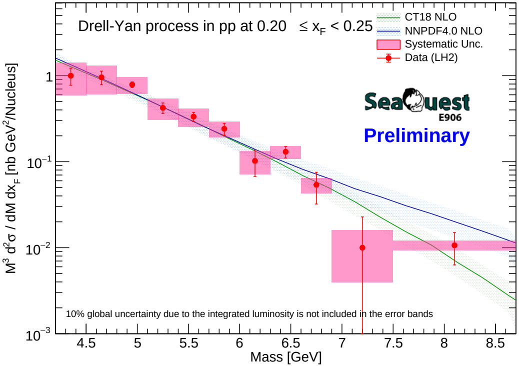
RS67



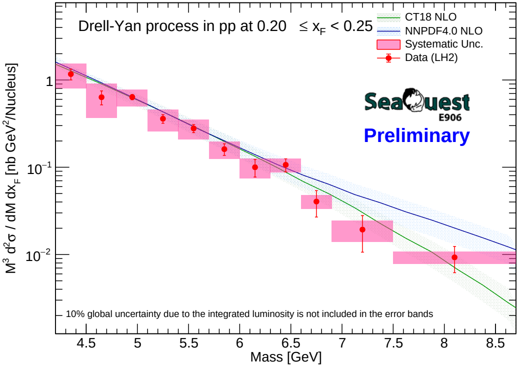
RS57-70



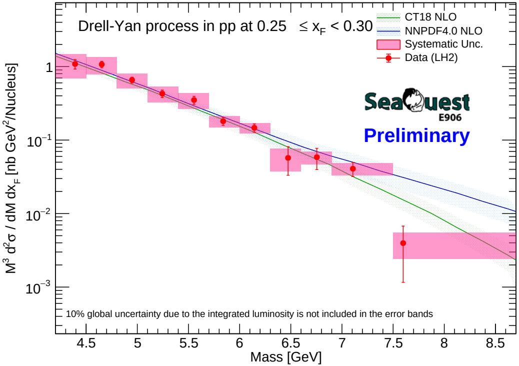
RS67


Drell-Yan process in pp at 0.20 ≤ x_F < 0.25
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\, d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

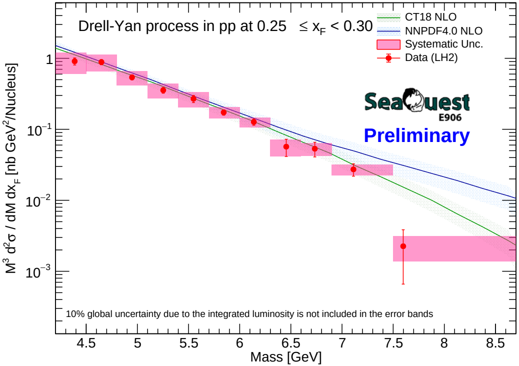
RS57-70


Drell-Yan process in pp at 0.20 ≤ x_F < 0.25
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\, d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS67

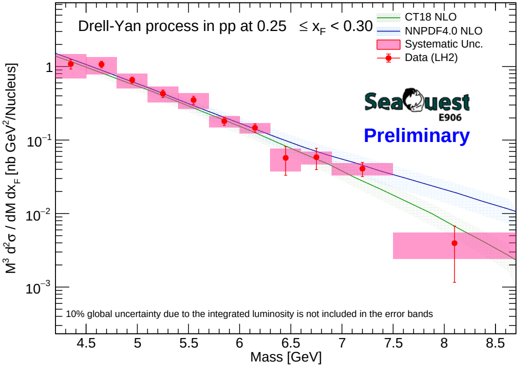

Drell-Yan process in pp at 0.25 ≤ x_F < 0.30
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70

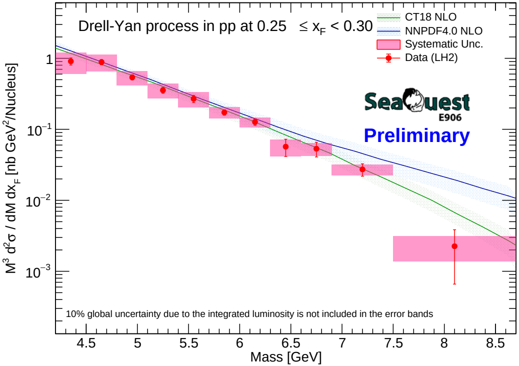

Drell-Yan process in pp at 0.25 ≤ x_F < 0.30
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

CrossSection_LH2_xF_0.25_0.30_GeoCenter_with_logo.pdf

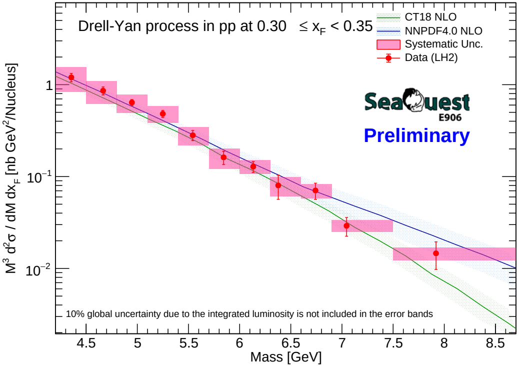
RS67



RS57-70

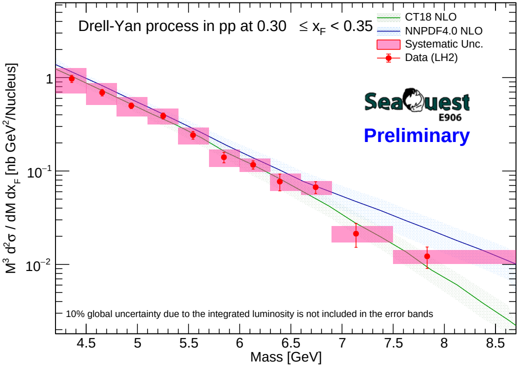


RS67



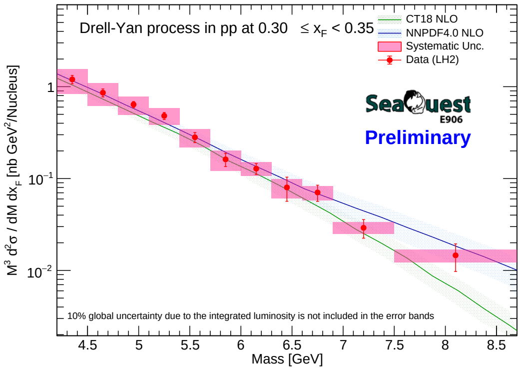
Drell-Yan process in pp at 0.30 ≤ x_F < 0.35
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\, d^2\sigma / dM\, dx_F$ [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70

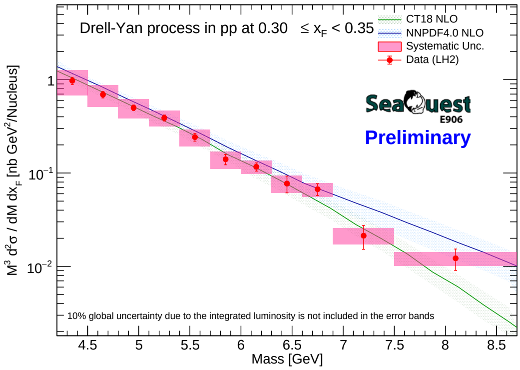


Drell-Yan process in pp at 0.30 ≤ x_F < 0.35
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\, d^2\sigma / dM\, dx_F$ [nb GeV^2/Nucleus]
Mass [GeV]

RS67

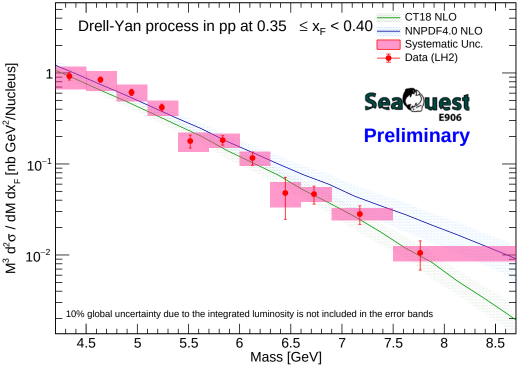

Drell-Yan process in pp at 0.30 ≤ x_F < 0.35
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70

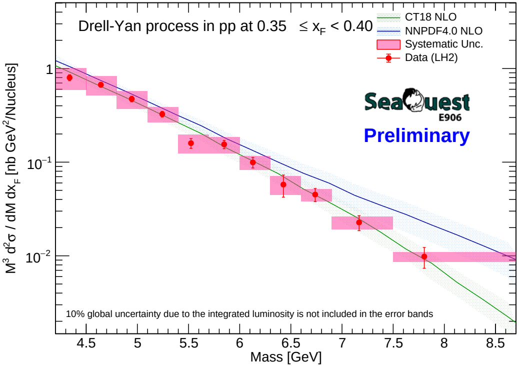

Drell-Yan process in pp at 0.30 ≤ x_F < 0.35
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

CrossSection_LH2_xF_0.35_0.40_Centroid_with_logo.pdf

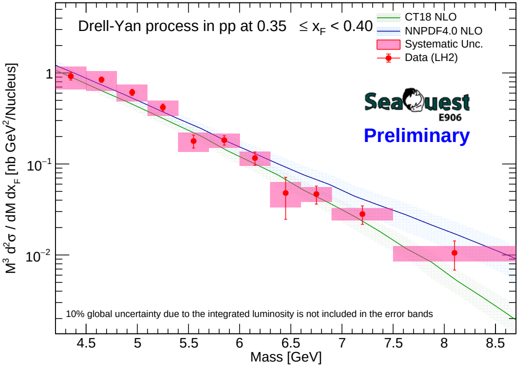
RS67



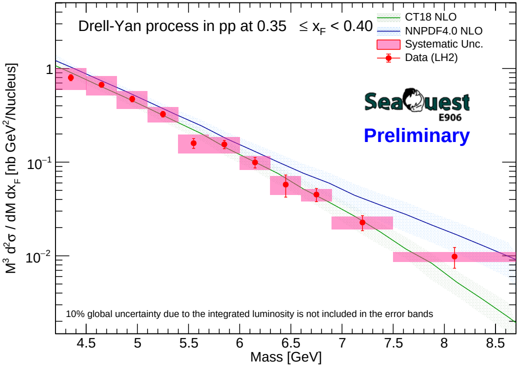
RS57-70



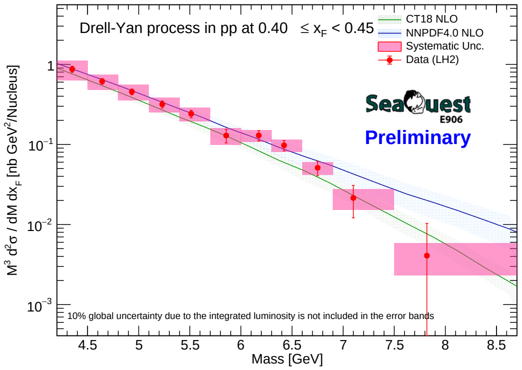
RS67


Drell-Yan process in pp at 0.35 ≤ x_F < 0.40
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

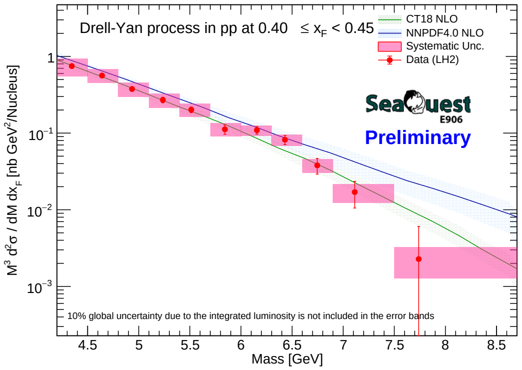
RS57-70


Drell-Yan process in pp at 0.35 ≤ x_F < 0.40
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS67

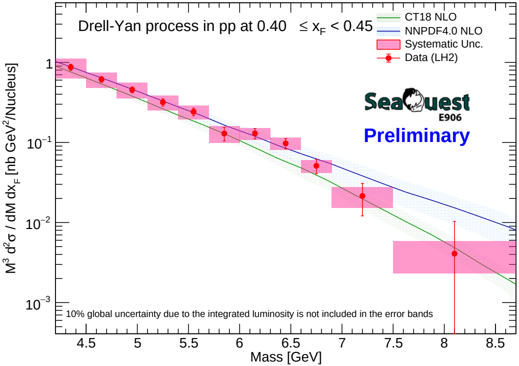

Drell-Yan process in pp at 0.40 ≤ x_F < 0.45
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70

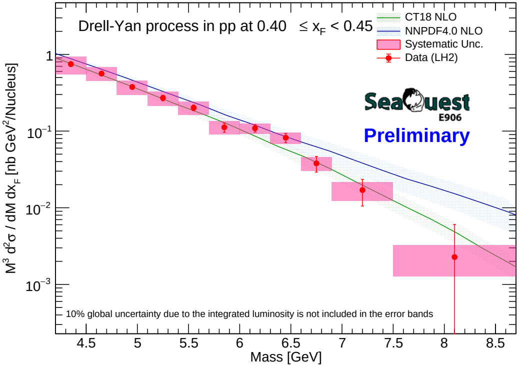

Drell-Yan process in pp at 0.40 ≤ x_F < 0.45
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

CrossSection_LH2_xF_0.40_0.45_GeoCenter_with_logo.pdf

RS67

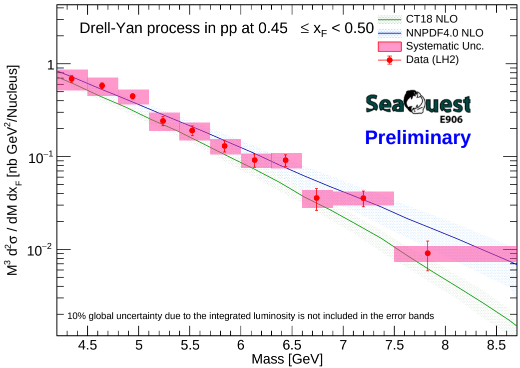


RS57-70

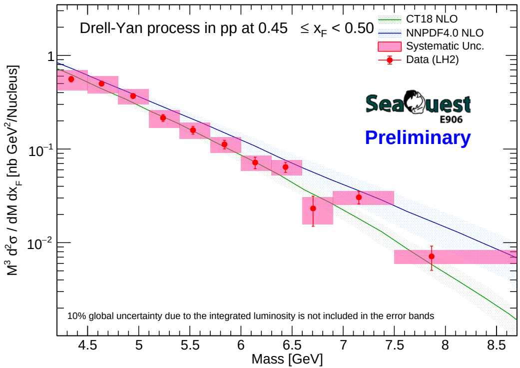


CrossSection_LH2_xF_0.45_0.50_Centroid_with_logo.pdf

RS67

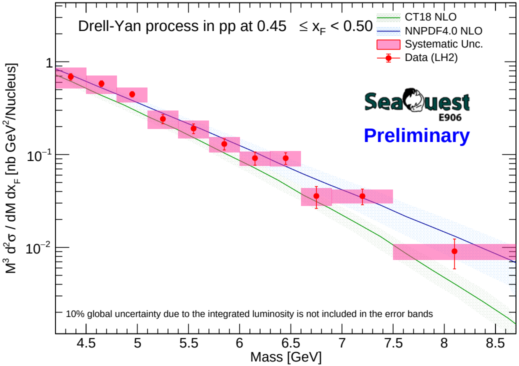


RS57-70

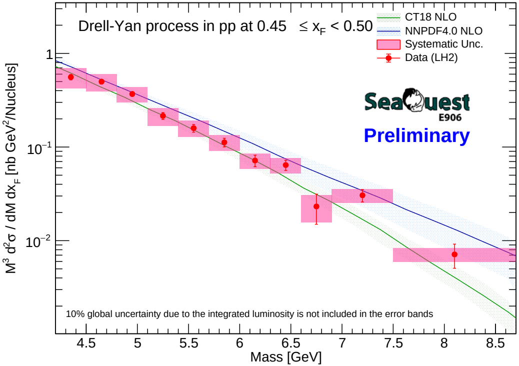


CrossSection_LH2_xF_0.45_0.50_GeoCenter_with_logo.pdf

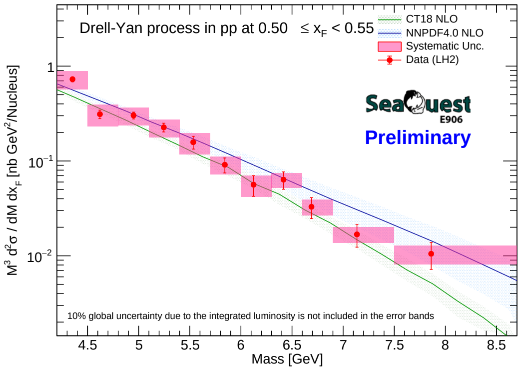
RS67



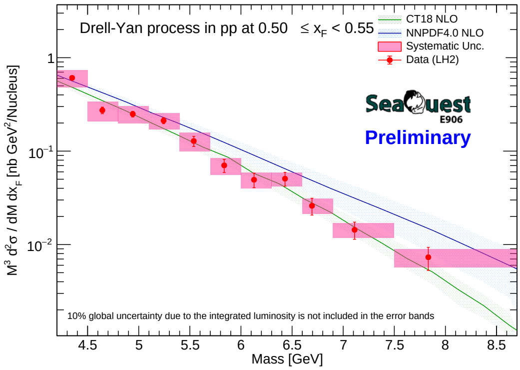
RS57-70



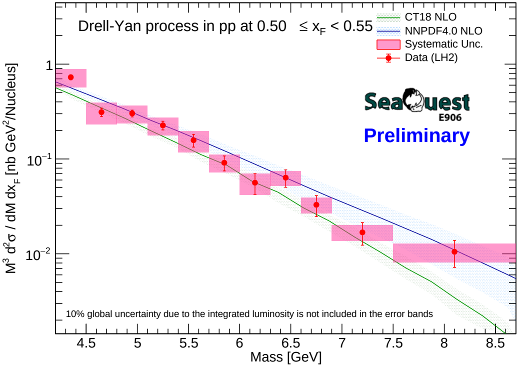
RS67


Drell-Yan process in pp at 0.50 ≤ x_F < 0.55
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

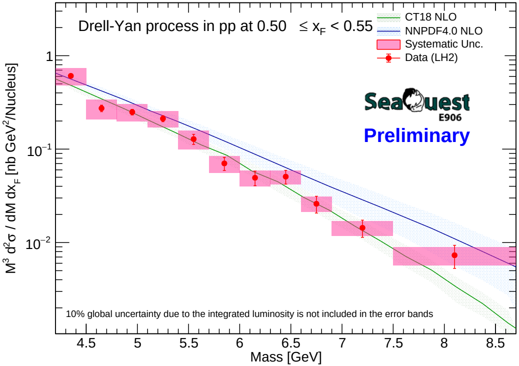
RS57-70


Drell-Yan process in pp at 0.50 ≤ x_F < 0.55
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

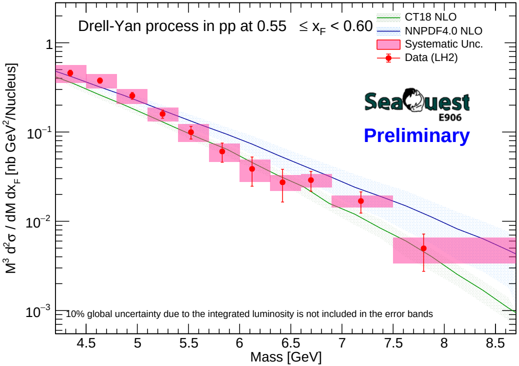
RS67


Drell-Yan process in pp at 0.50 ≤ x_F < 0.55
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

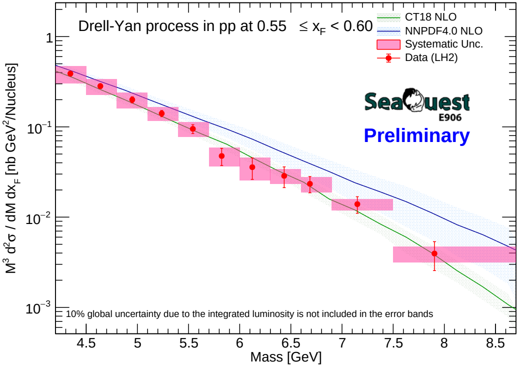
RS57-70


Drell-Yan process in pp at 0.50 ≤ x_F < 0.55
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS67

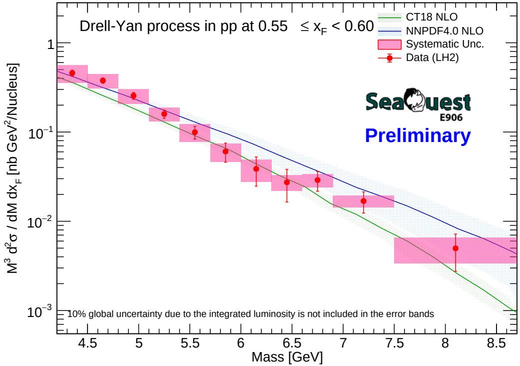

Drell-Yan process in pp at 0.55 ≤ x_F < 0.60
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70

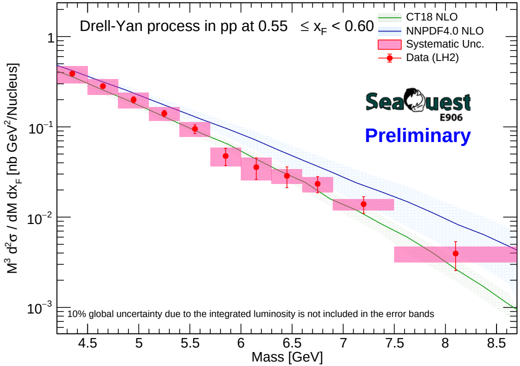

Drell-Yan process in pp at 0.55 ≤ x_F < 0.60
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

CrossSection_LH2_xF_0.55_0.60_GeoCenter_with_logo.pdf

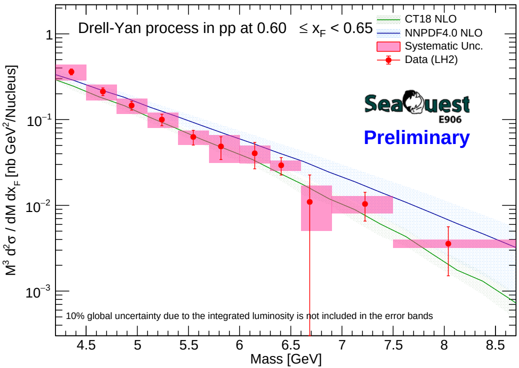
RS67



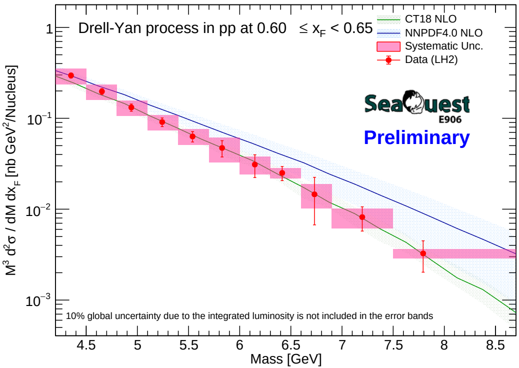
RS57-70



RS67

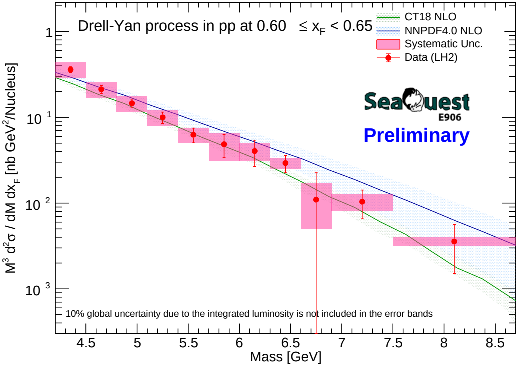

Drell-Yan process in pp at 0.60 ≤ x_F < 0.65
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70

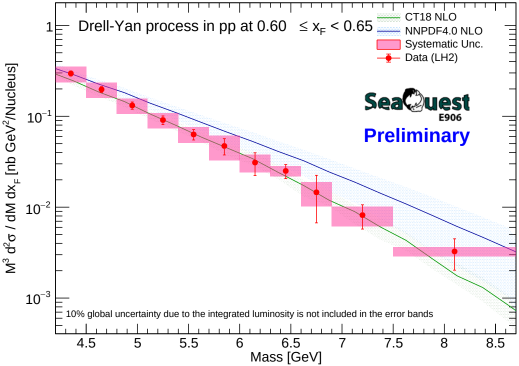

Drell-Yan process in pp at 0.60 ≤ x_F < 0.65
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

CrossSection_LH2_xF_0.60_0.65_GeoCenter_with_logo.pdf

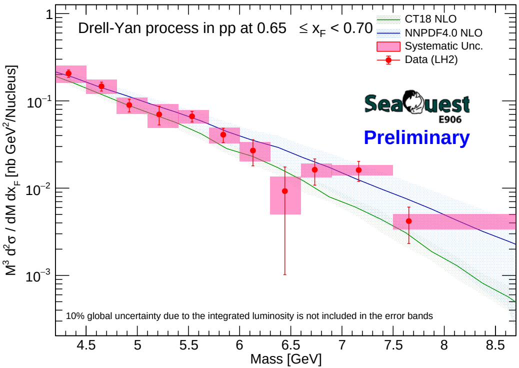
RS67



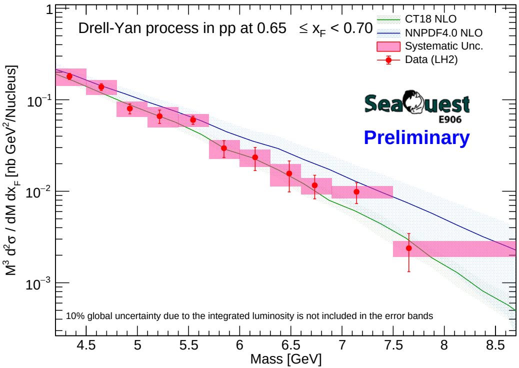
RS57-70



RS67

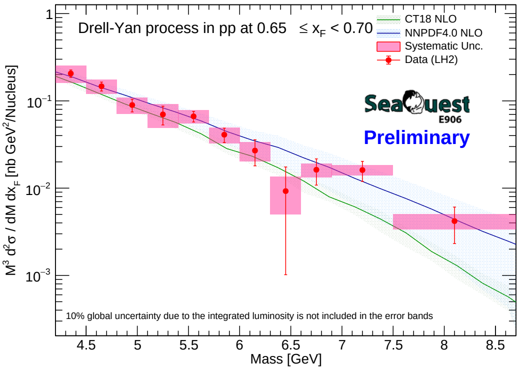

Drell-Yan process in pp at 0.65 ≤ x_F < 0.70
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70

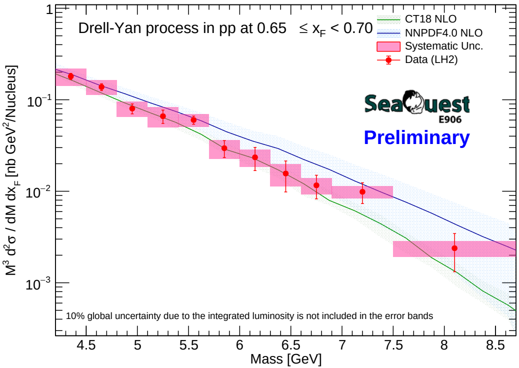

Drell-Yan process in pp at 0.65 ≤ x_F < 0.70
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest
E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\ d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

CrossSection_LH2_xF_0.65_0.70_GeoCenter_with_logo.pdf

RS67

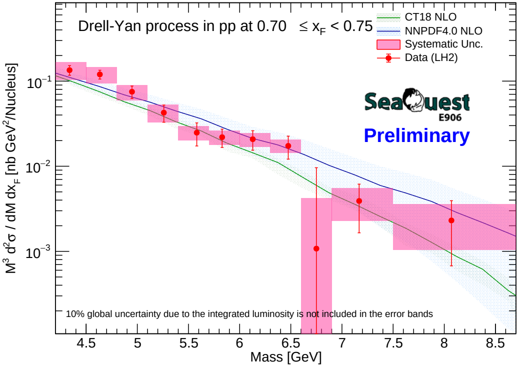


RS57-70

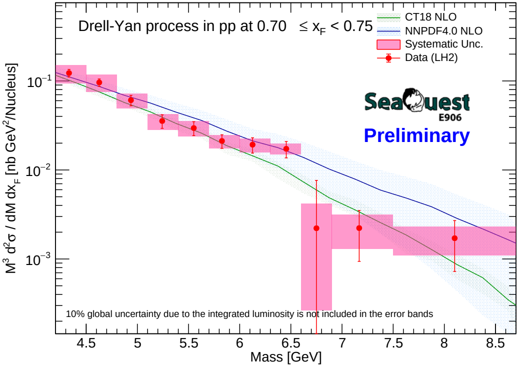


CrossSection_LH2_xF_0.70_0.75_Centroid_with_logo.pdf

RS67

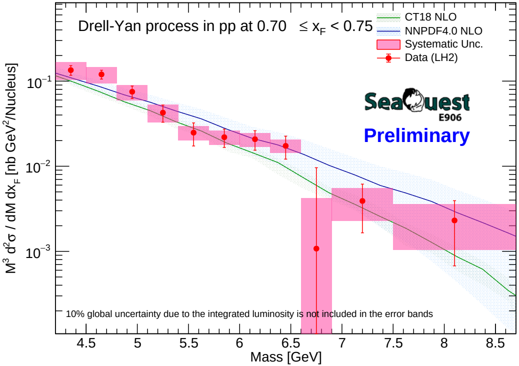


RS57-70



CrossSection_LH2_xF_0.70_0.75_GeoCenter_with_logo.pdf

RS67



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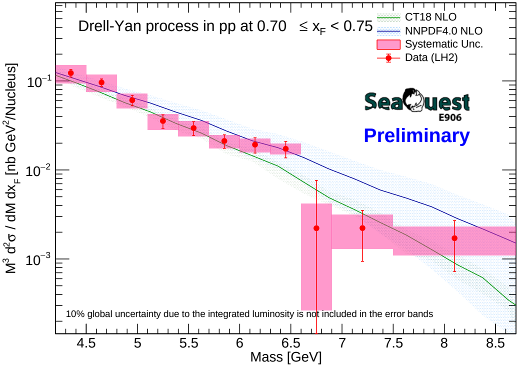
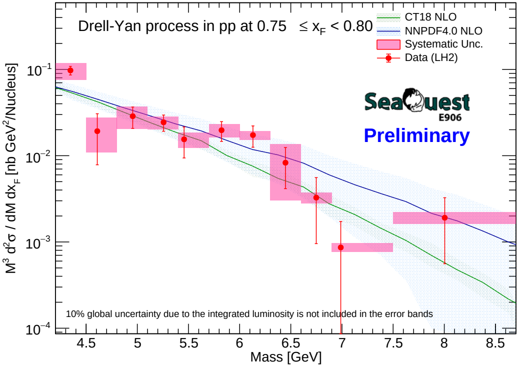
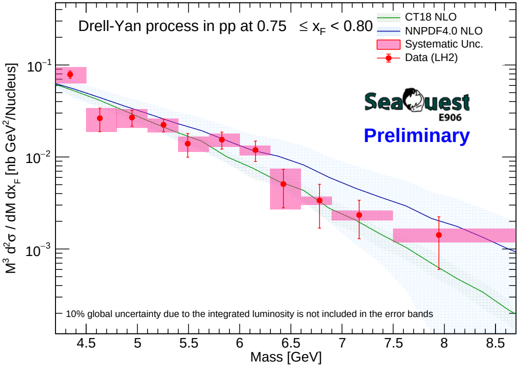


RS67

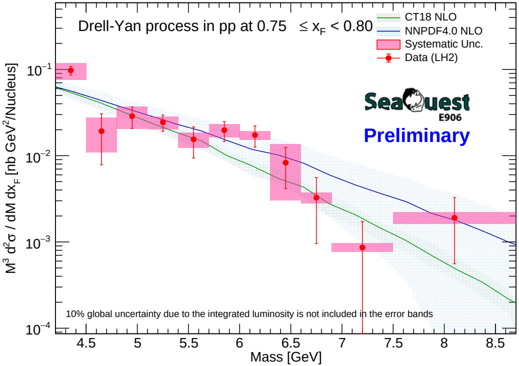

Drell-Yan process in pp at 0.75 ≤ x_F < 0.80
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\, d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

RS57-70

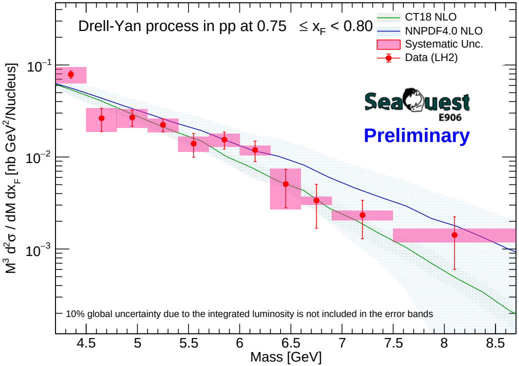

Drell-Yan process in pp at 0.75 ≤ x_F < 0.80
CT18 NLO
NNPDF4.0 NLO
Systematic Unc.
Data (LH2)
SeaQuest E906
Preliminary
10% global uncertainty due to the integrated luminosity is not included in the error bands
$M^3\, d^2\sigma$ / dM dx_F [nb GeV^2/Nucleus]
Mass [GeV]

CrossSection_LH2_xF_0.75_0.80_GeoCenter_with_logo.pdf

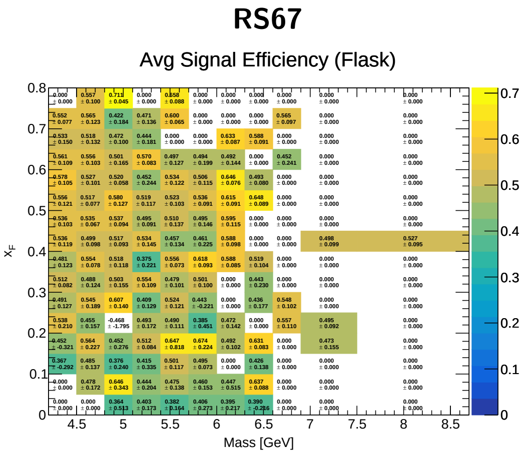
RS67

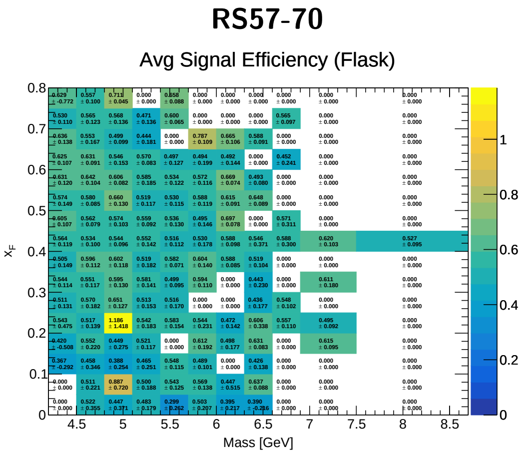


RS57-70



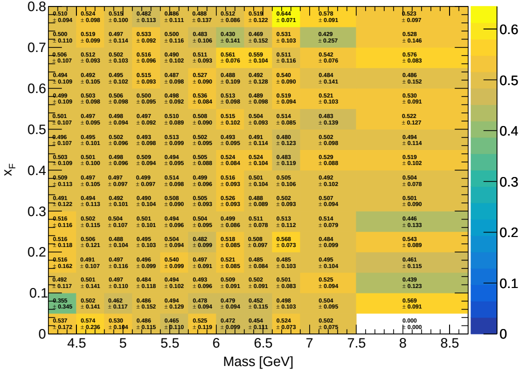
E_final_signal_Flask.pdf



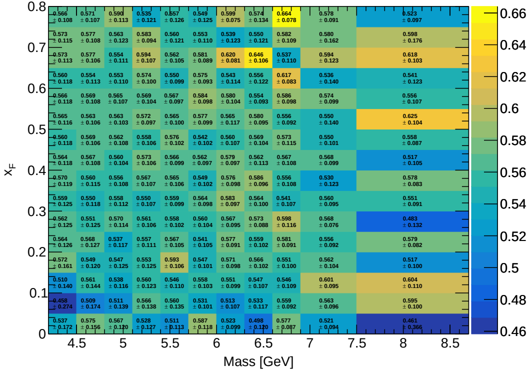


E_final_signal_LD2.pdf

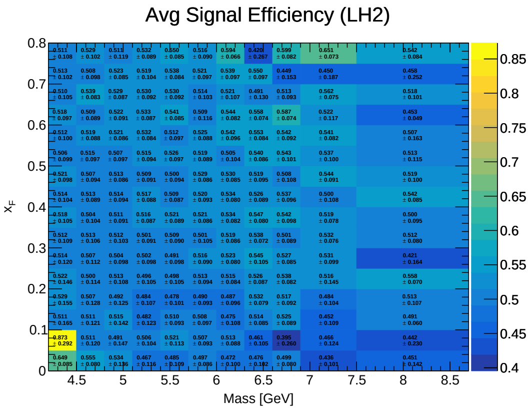
RS67



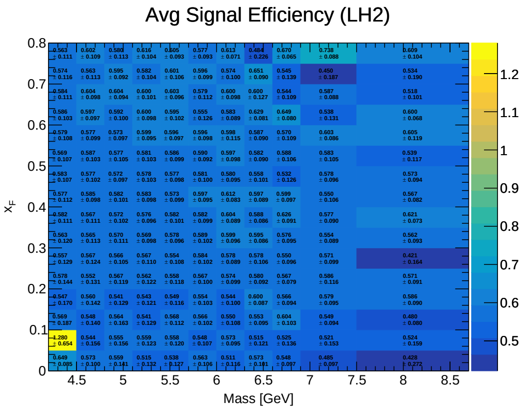
RS57-70



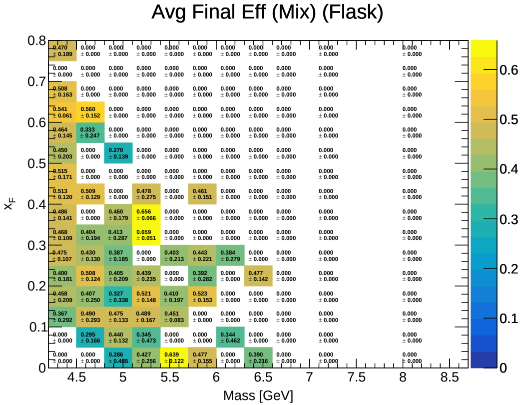
RS67


Avg Signal Efficiency (LH2)
x_F
Mass [GeV]

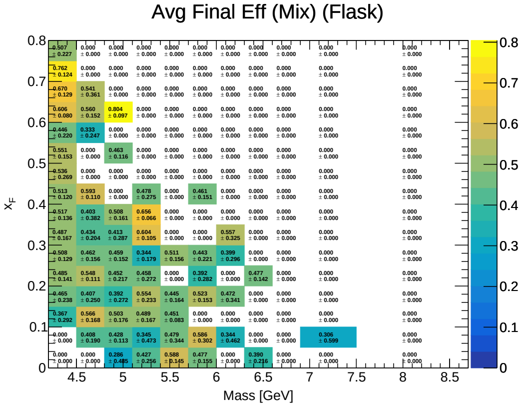
RS57-70


Avg Signal Efficiency (LH2)
x_F
Mass [GeV]

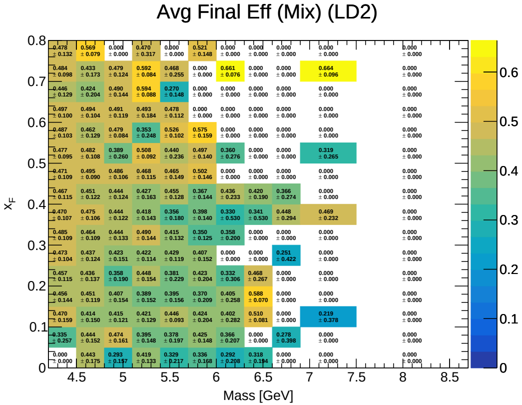
RS67


Avg Final Eff (Mix) (Flask)
x_F
Mass [GeV]
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
4.5
5
5.5
6
6.5
7
7.5
8
8.5

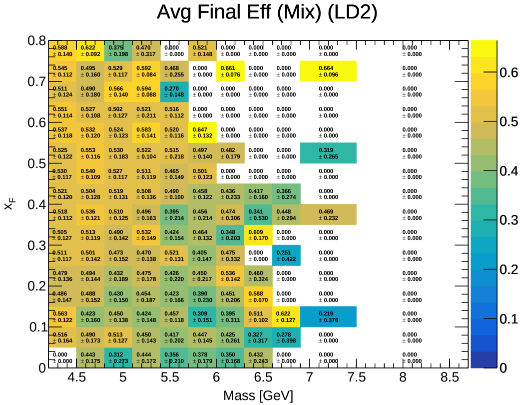
RS57-70


Avg Final Eff (Mix) (Flask)
x_F
Mass [GeV]
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
4.5
5
5.5
6
6.5
7
7.5
8
8.5

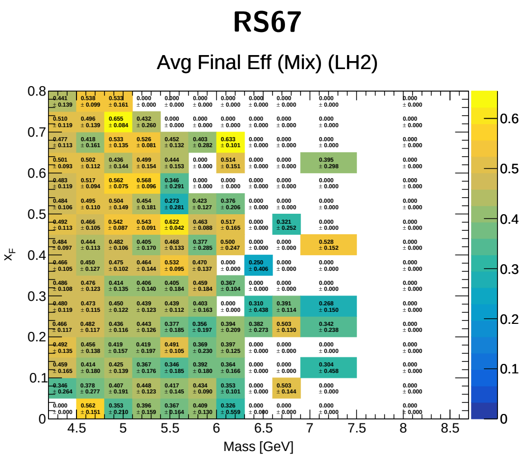
RS67

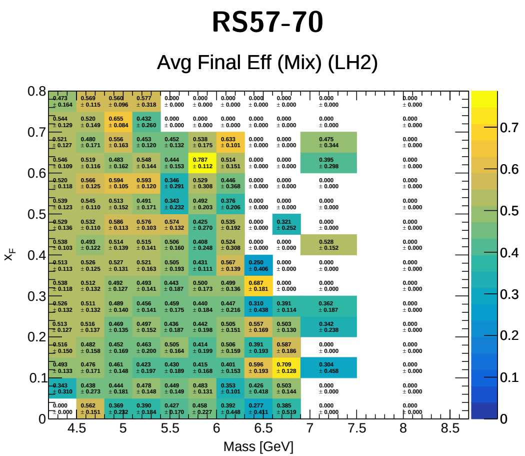

Avg Final Eff (Mix) (LD2)
x_F
Mass [GeV]

RS57-70

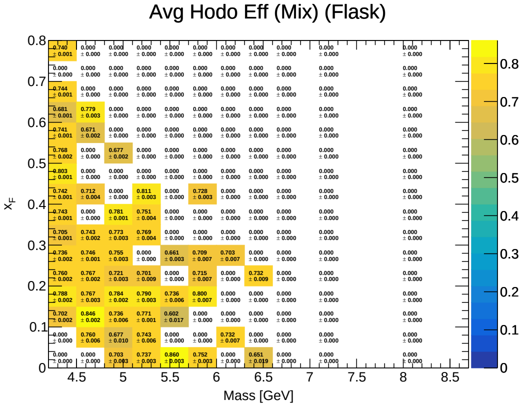

Avg Final Eff (Mix) (LD2)
x_F
Mass [GeV]

E_mix_final_LH2.pdf

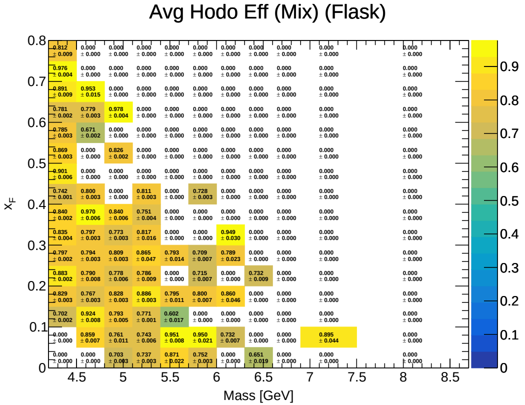




RS67

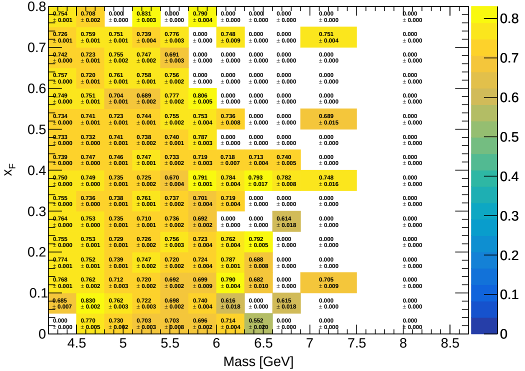

Avg Hodo Eff (Mix) (Flask)
Mass [GeV]
x_F

RS57-70


Avg Hodo Eff (Mix) (Flask)
Mass [GeV]
x_F

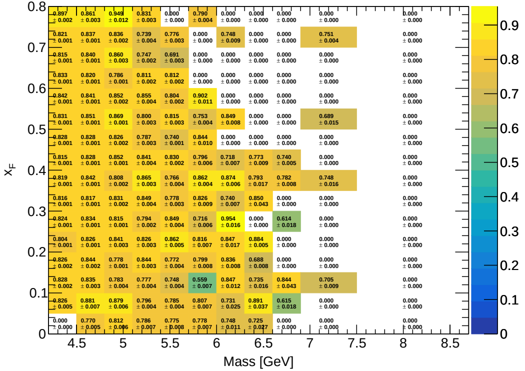
RS67

Avg Hodo Eff (Mix) (LD2)



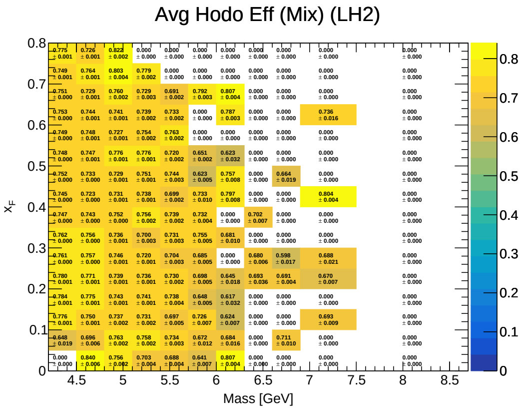
RS57-70

Avg Hodo Eff (Mix) (LD2)

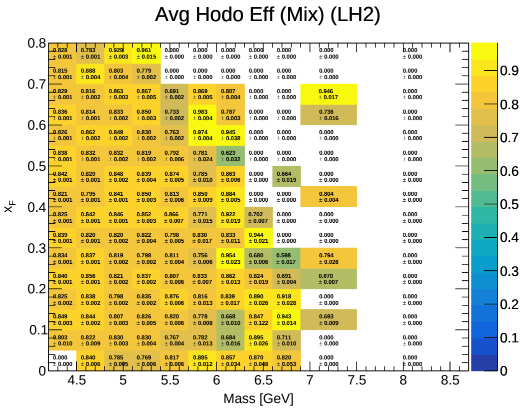


E_mix_hodo_LH2.pdf

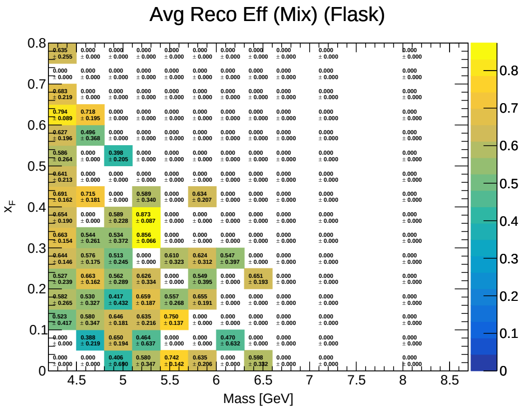
RS67



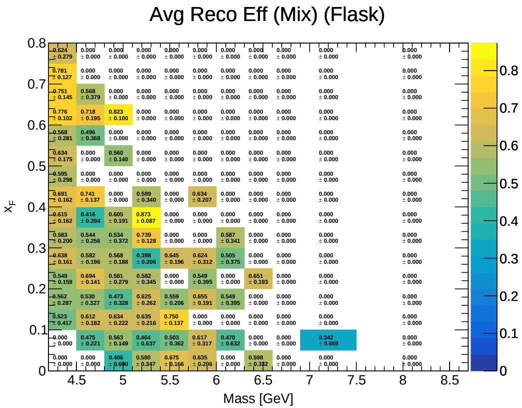
RS57-70



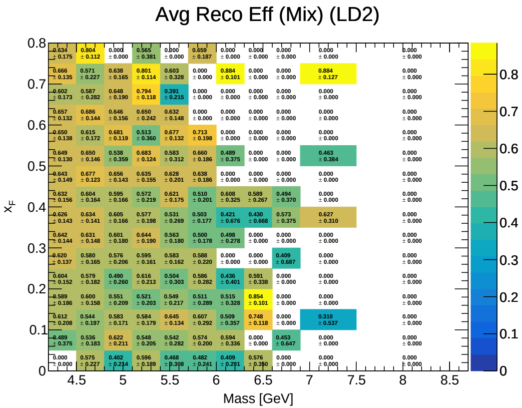
RS67


Avg Reco Eff (Mix) (Flask)
x_F
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
4.5
5
5.5
6
6.5
7
7.5
8
8.5
Mass [GeV]

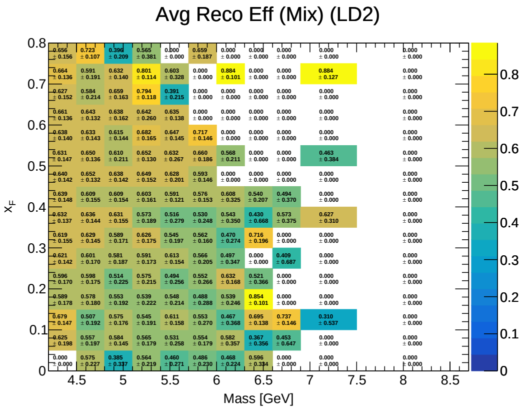
RS57-70


Avg Reco Eff (Mix) (Flask)
x_F
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
4.5
5
5.5
6
6.5
7
7.5
8
8.5
Mass [GeV]

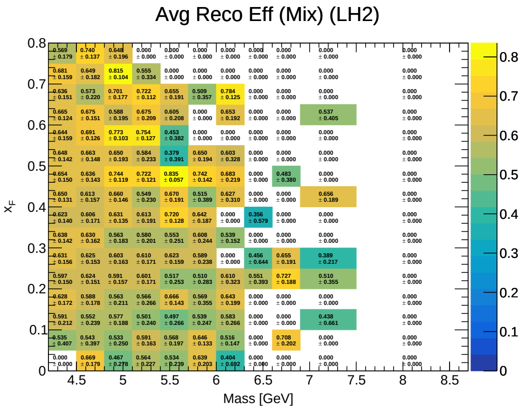
RS67


Avg Reco Eff (Mix) (LD2)
x_F
Mass [GeV]

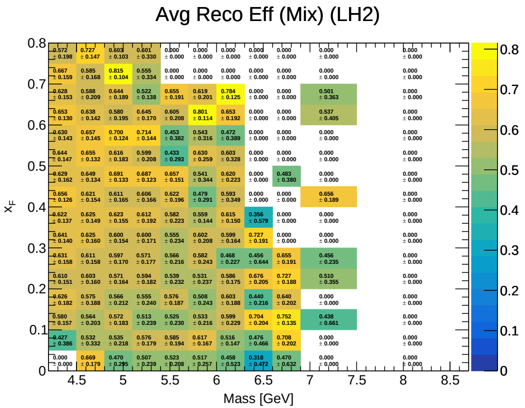
RS57-70


Avg Reco Eff (Mix) (LD2)
x_F
Mass [GeV]

RS67

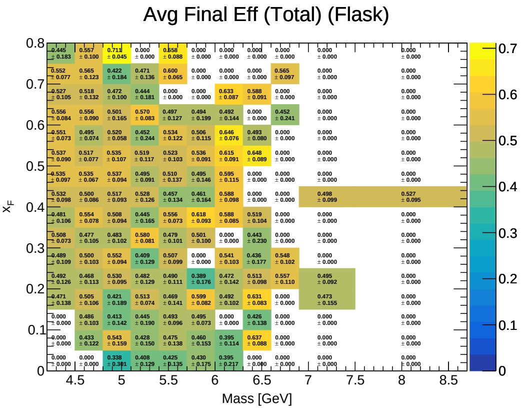

Avg Reco Eff (Mix) (LH2)
x_F
Mass [GeV]
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
4.5
5
5.5
6
6.5
7
7.5
8
8.5

RS57-70

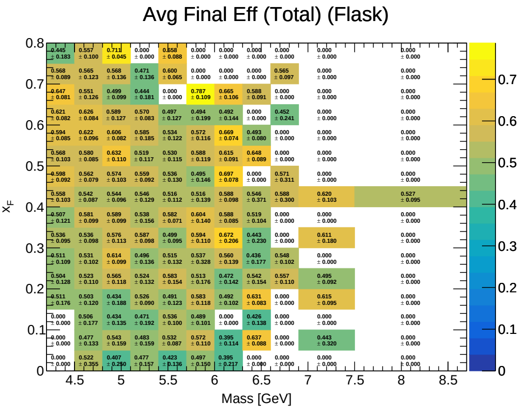

Avg Reco Eff (Mix) (LH2)
x_F
Mass [GeV]
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
4.5
5
5.5
6
6.5
7
7.5
8
8.5

E_total_final_Flask.pdf

RS67

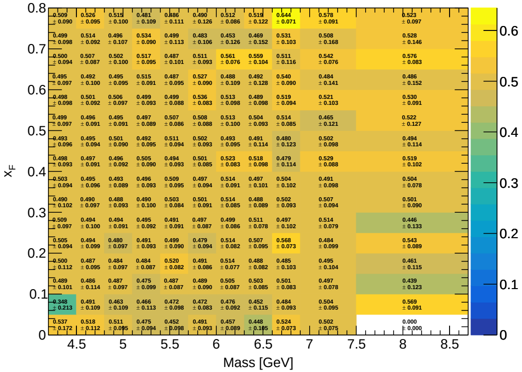


RS57-70

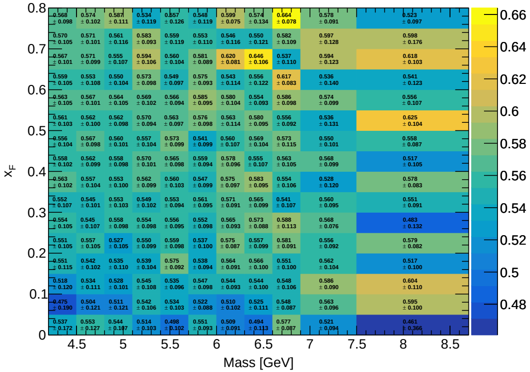


E_total_final_LD2.pdf

RS67

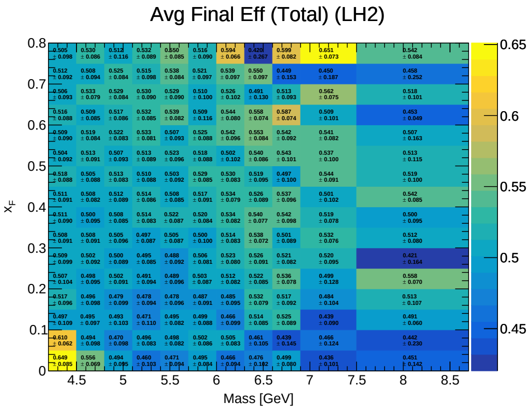


RS57-70

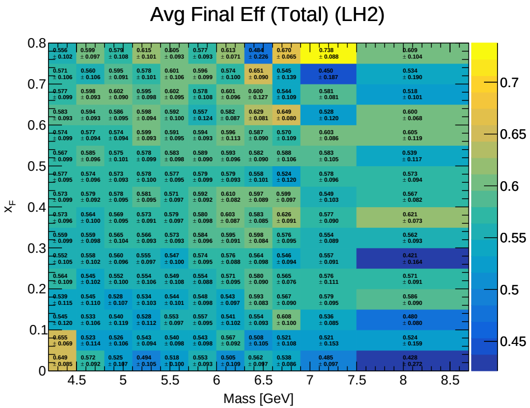


E_total_final_LH2.pdf

RS67

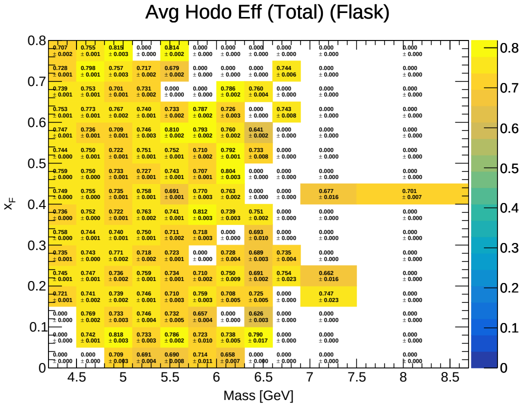


RS57-70

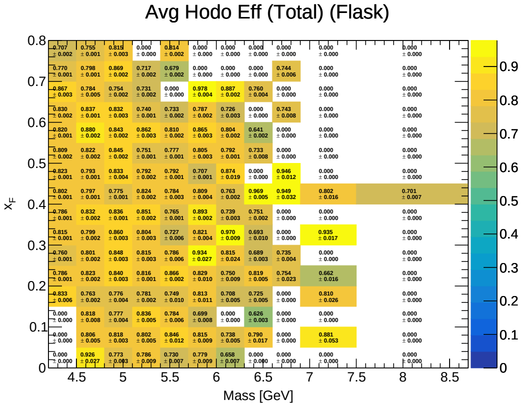


E_total_hodo_Flask.pdf

RS67

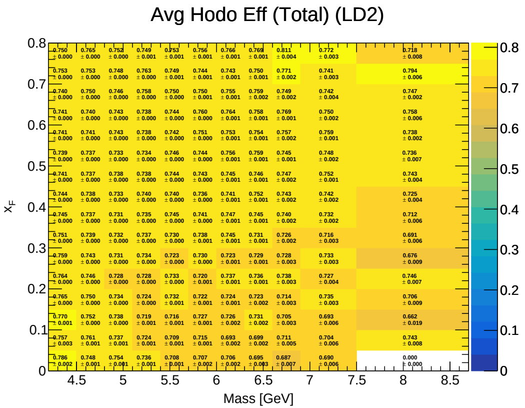


RS57-70

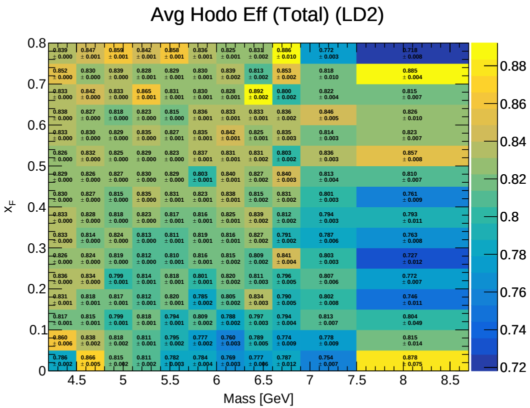


E_total_hodo_LD2.pdf

RS67

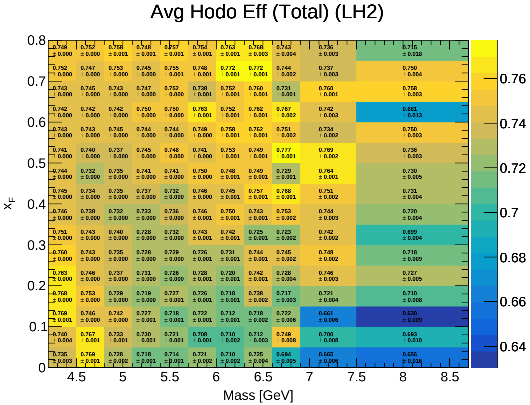


RS57-70

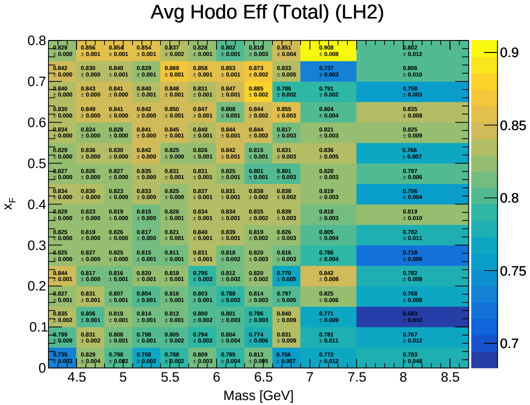


E_total_hodo_LH2.pdf

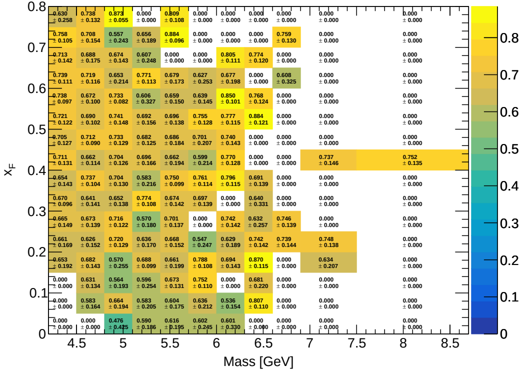
RS67



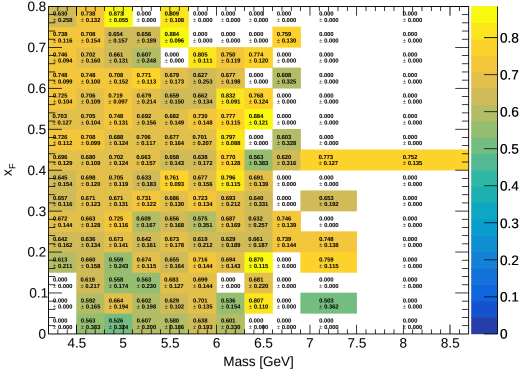
RS57-70



RS67

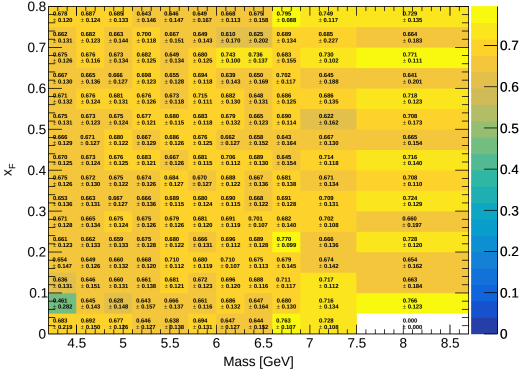

Avg Reco Eff (Total) (Flask)
x_F
Mass [GeV]

RS57-70

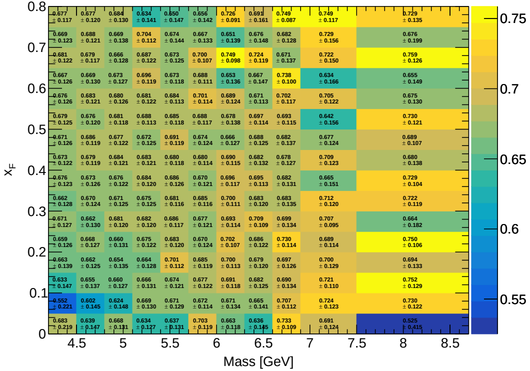

Avg Reco Eff (Total) (Flask)
x_F
Mass [GeV]

E_total_reco_LD2.pdf

RS67

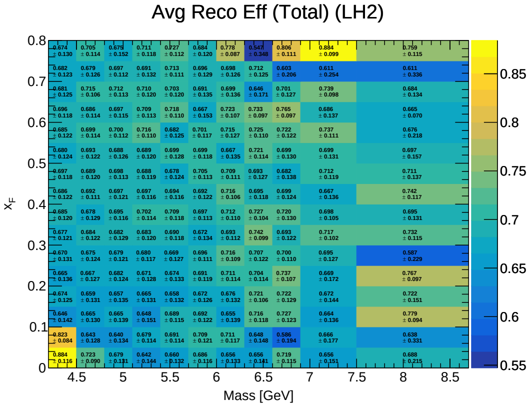


RS57-70

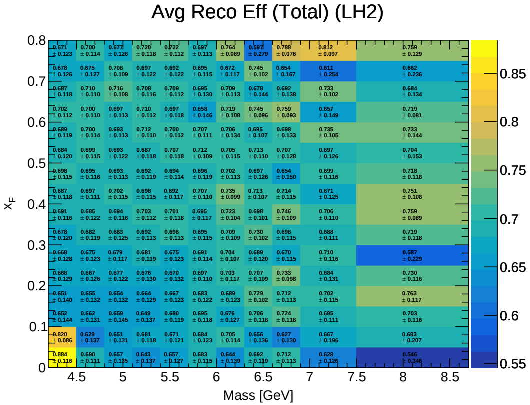


E_total_reco_LH2.pdf

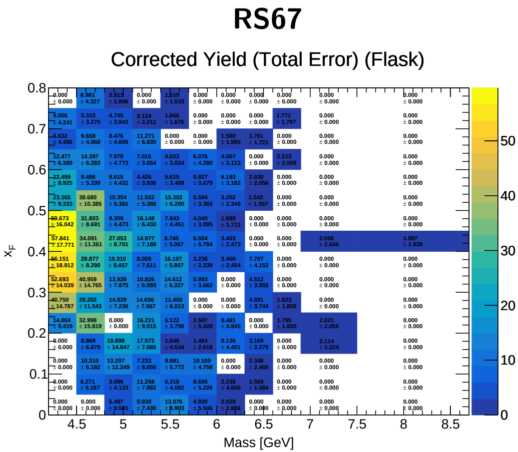
RS67

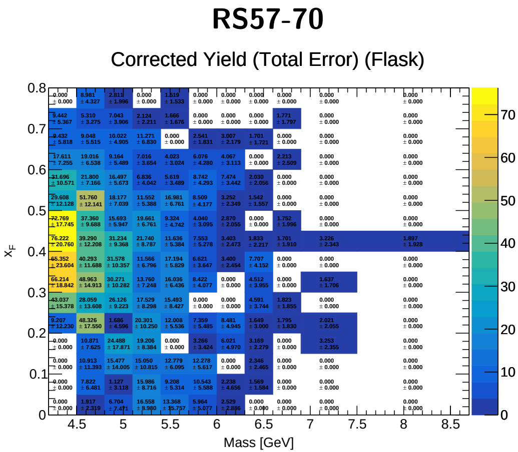


RS57-70



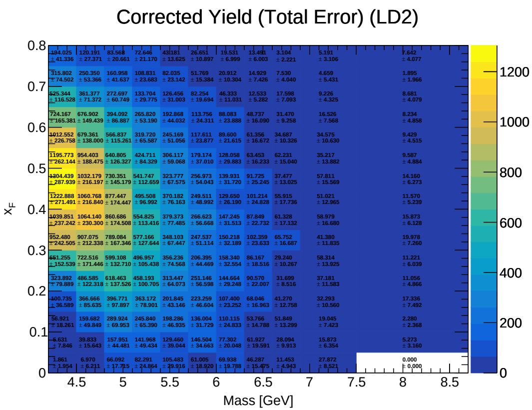
Y_corrected_Flask.pdf



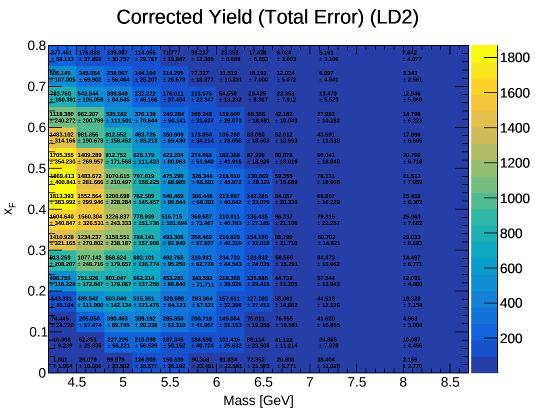


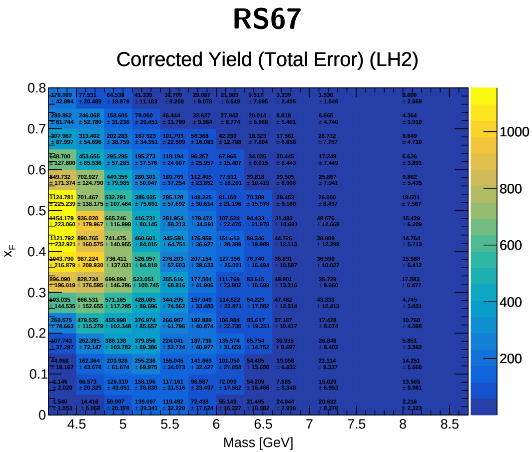
Y_corrected_LD2.pdf

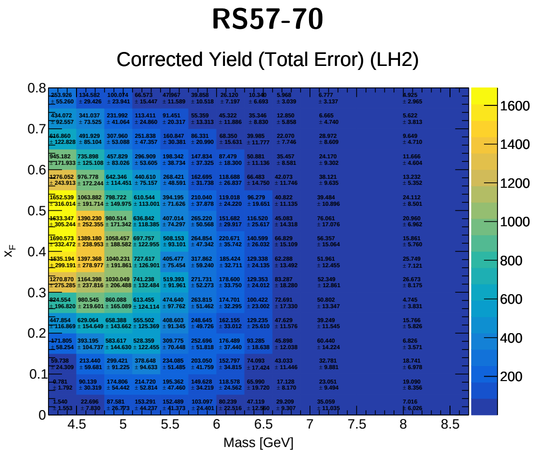
RS67

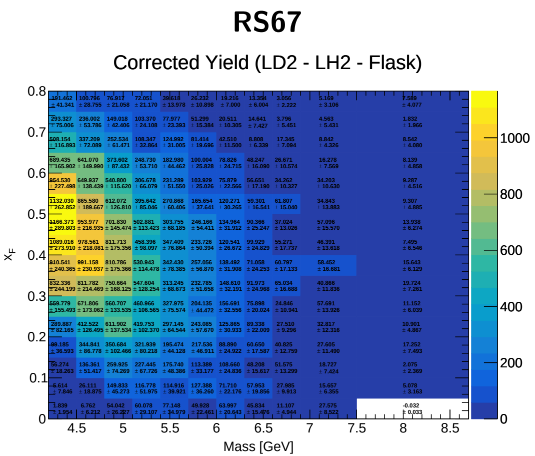


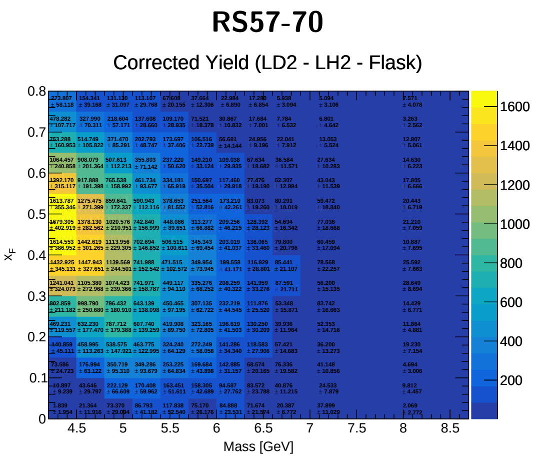
RS57-70



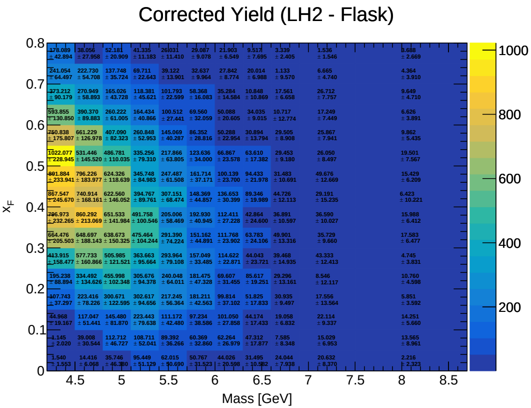

RS67
Corrected Yield (Total Error) (LH2)
x_F
Mass [GeV]
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
4.5
5
5.5
6
6.5
7
7.5
8
8.5
1000
800
600
400
200


RS57-70
Corrected Yield (Total Error) (LH2)
x_F
Mass [GeV]
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
4.5
5
5.5
6
6.5
7
7.5
8
8.5
1600
1400
1200
1000
800
600
400
200

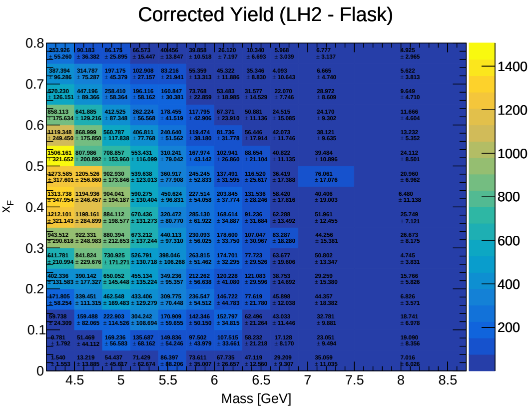

RS67
Corrected Yield (LD2 - LH2 - Flask)
x_F
Mass [GeV]
0
200
400
600
800
1000


RS57-70
Corrected Yield (LD2 - LH2 - Flask)
x_F
Mass [GeV]
200
400
600
800
1000
1200
1400
1600

RS67

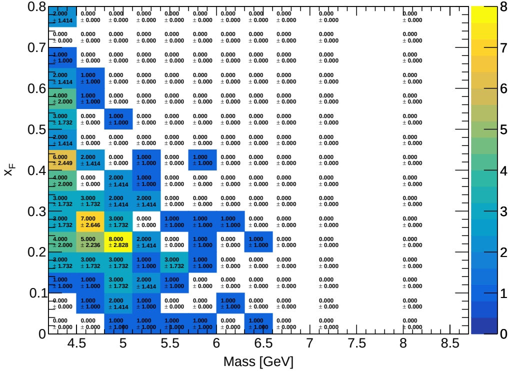

Corrected Yield (LH2 - Flask)
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
x_F
4.5
5
5.5
6
6.5
7
7.5
8
8.5
Mass [GeV]
1000
800
600
400
200

RS57-70


Corrected Yield (LH2 - Flask)
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
x_F
4.5
5
5.5
6
6.5
7
7.5
8
8.5
Mass [GeV]
1400
1200
1000
800
600
400
200

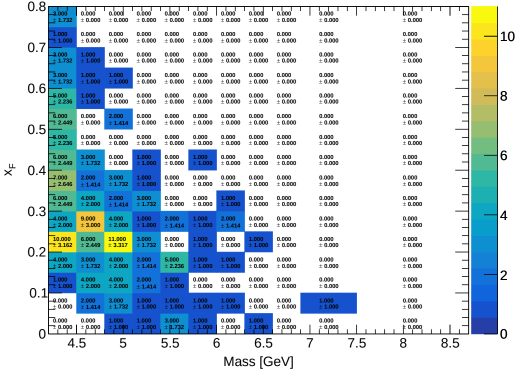
RS67

Mix Yield (result_mix) (Flask)



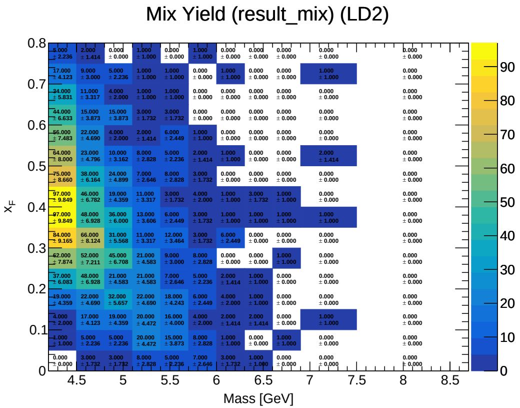
RS57-70

Mix Yield (result_mix) (Flask)

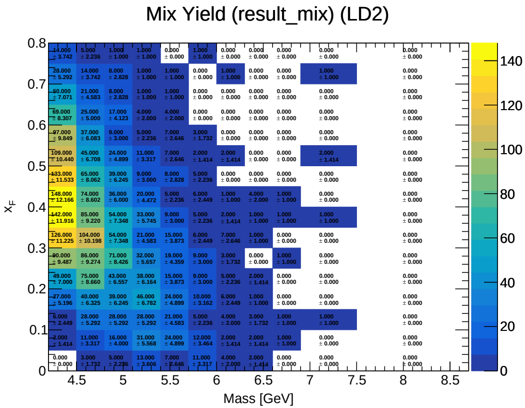


Y_mix_LD2.pdf

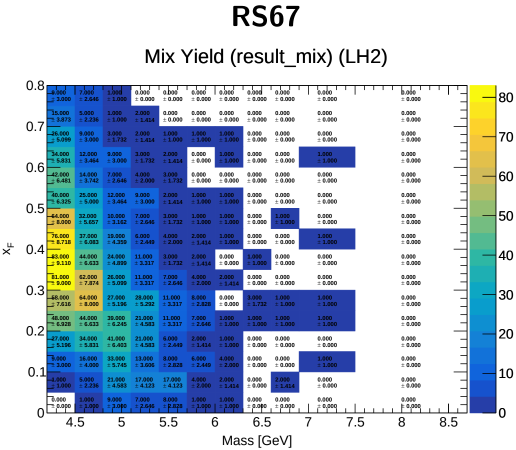
RS67

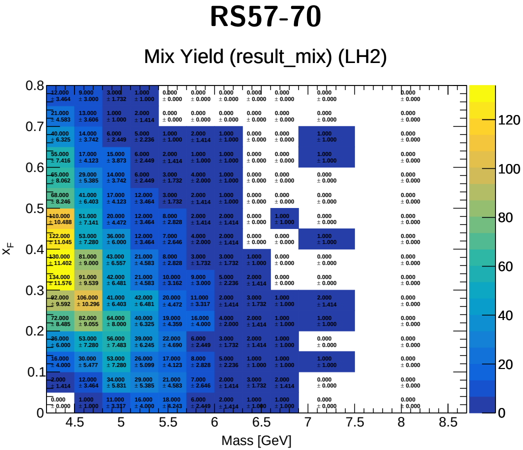


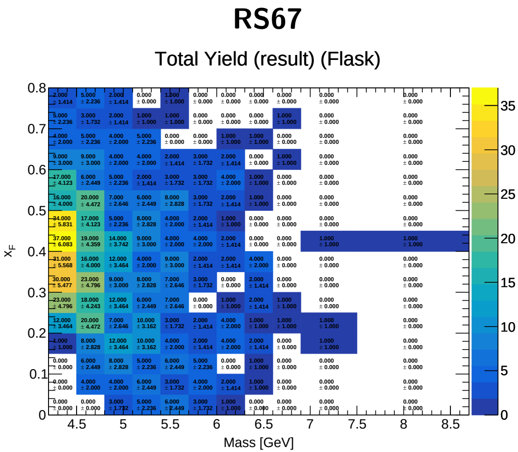
RS57-70

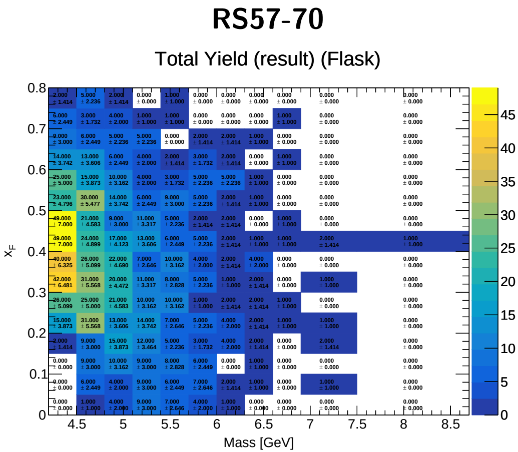


Y_mix_LH2.pdf


RS67
Mix Yield (result_mix) (LH2)
x_F
Mass [GeV]

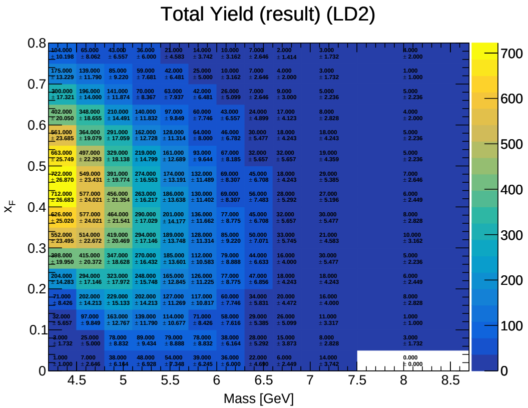

RS57-70
Mix Yield (result_mix) (LH2)
x_F
Mass [GeV]


RS67
Total Yield (result) (Flask)
x_F
Mass [GeV]
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8
4.5 5 5.5 6 6.5 7 7.5 8 8.5
0 5 10 15 20 25 30 35

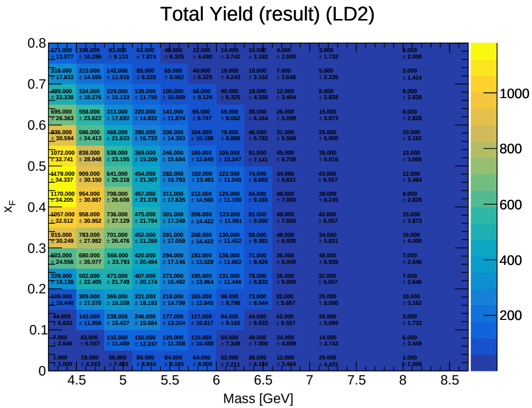

RS57-70
Total Yield (result) (Flask)
x_F
Mass [GeV]
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8
4.5 5 5.5 6 6.5 7 7.5 8 8.5
0 5 10 15 20 25 30 35 40 45

Y_total_LD2.pdf

RS67

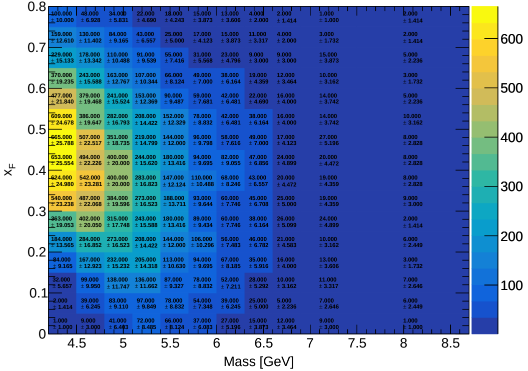


RS57-70



RS67

Total Yield (result) (LH2)



RS57-70

Total Yield (result) (LH2)